

MethylPipeline Usage Manual

Operational runbook for DomainProgram workflows and staged validation

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Table of contents

1	MethylPipeline Usage Manual	8
1.1	What this manual is	8
1.2	How to run staged validation	8
1.3	Reading paths	8
1.4	End-to-end workflow map	9
1.5	Scope boundaries	9
1.6	Canonical references	9
2	Relationship to Theory and Implementation	10
I	Part I: Setup and Package Orientation	11
3	Prerequisites and Environment	12
3.1	Purpose	12
3.2	Required tools	12
3.3	Environment bootstrap	12
3.4	Verify CLI availability	12
3.5	Runtime assumptions	13
3.6	Success checks	13
3.7	Common failures and fixes	13
3.8	See also	13
4	Project Config and Layout	14
4.1	Purpose	14
4.2	Four-layer configuration	14
4.3	Repository vs /work (what goes where)	14
4.4	Minimum study manifest	15
4.5	Validation / Monte Carlo keys (profile <code>actionConfig.validation</code>)	15
4.6	Output layout	16
4.7	Handoff model	16
4.8	Do not do this	17
4.9	Success checks	17
4.10	See also	17
5	Sample Prep and Quality Control	18
5.1	Purpose	18
5.2	Alignment modes (linear vs pangenome)	18
5.3	End-to-end flow	21
5.4	Where artifacts live	22

5.5	Alignment QC: inputs and normalized JSON	22
5.5.1	Picard-style duplication metrics	22
5.5.2	Parabricks WGBS metrics JSON	23
5.5.3	Our equivalent export JSON (V2)	23
5.6	Alignment guardrails: what “good alignment” means	23
5.6.1	Optional and analyte-specific checks	24
5.7	Cycle screening and automated recovery	25
5.8	Extraction QC: confirming enough good reads	25
5.8.1	Input: extraction manifest	26
5.8.2	Output: extraction QC JSON	26
5.8.3	Why both gates matter	26
5.9	Pass, fail, and handoff	26
5.10	Configuration checklist	27
5.11	Commands (standalone debugging)	27
5.12	Related documentation	28
6	Orchestration: methyl-workflow-run (canonical)	29
6.1	Purpose	29
6.2	Preferred vs legacy	29
6.3	What you pass	29
6.4	Programs by scope	30
6.5	Local development without a database	30
6.6	Legacy methyl-validation (when it still appears)	31
6.7	Related	31
7	Package Reference (Individual Usage)	32
7.1	Purpose	32
7.2	Package quick map	32
7.3	Standalone examples	32
7.3.1	Alignment QC	32
7.3.2	Enrichment from mapper output	33
7.3.3	Blind prediction with frozen model	33
7.4	Orchestrated equivalents in methyl-validation	33
7.5	Integration diagram	33
7.6	Do not do this	33
7.7	See also	34
II	Part II: Staged Solution Workflow	35
8	Stage: Stability	36
8.1	Purpose	36
8.2	Entry command (canonical)	36
8.3	Legacy entry command	36
8.4	Required config keys	37
8.5	Statistical modeling modes (generic profiles)	37
8.6	Process model	38
8.7	Expected outputs	38

8.8	Success checks	38
8.9	Do not do this	38
8.10	Recovery	39
8.11	See also	39
9	Stage: Freeze	40
9.1	Purpose	40
9.2	Entry command (canonical)	40
9.3	Legacy entry command	40
9.4	Prerequisites	40
9.5	Process model	41
9.6	Expected outputs	41
9.7	Success checks	41
9.8	Recommended post-freeze gate	41
9.9	Do not do this	42
9.10	Recovery	42
9.11	See also	42
10	Stage: Model	43
10.1	Purpose	43
10.2	Common execution patterns	43
10.2.1	Pattern A: Full lifecycle (canonical)	43
10.2.2	Pattern B: Direct production model build (legacy CLI)	43
10.2.3	Pattern C: Backend model-MC comparison then select best (legacy CLI)	43
10.3	Backend behavior	44
10.4	Process model	44
10.5	Required config keys	44
10.6	Expected outputs	45
10.7	Success checks	45
10.8	Do not do this	45
10.9	See also	45
11	Stage: Post-model Validation	46
11.1	Purpose	46
11.2	Entry command (canonical)	46
11.3	Legacy entry command	46
11.4	Prerequisites	46
11.5	Process model	47
11.6	Expected outputs	47
11.7	Success checks	47
11.8	Do not do this	47
11.9	Recovery	47
11.10	See also	47
12	Stage: Blind Prediction	48
12.1	Purpose	48
12.2	Important boundary	48
12.3	Entry command (standalone)	48

12.4	Process model	48
12.5	Required config pattern	48
12.6	Expected outputs	48
12.7	Success checks	49
12.8	Do not do this	49
12.9	Recovery	49
12.10	See also	49
III	Part III: Operations and Reliability	50
13	Artifacts and QA Checks	51
13.1	Purpose	51
13.2	Workflow action observability	51
13.3	Artifact map	51
13.4	Handoff integrity checks	52
13.5	QA checklist per stage	52
13.5.1	Stability	52
13.5.2	Freeze	52
13.5.3	Model	52
13.5.4	Post-model validation	53
13.5.5	Blind prediction	53
13.6	Escalation triggers	53
13.7	See also	53
14	Troubleshooting and Recovery	54
14.1	Purpose	54
14.2	Failure routing	54
14.3	Common problems	54
14.4	Safe recovery sequence	55
14.5	Resume examples	55
14.6	Operational guardrails	55
14.7	See also	55
15	Command Cookbook	56
15.1	Purpose	56
15.2	1) Full staged run (canonical — DomainProgram)	56
15.3	2) Full staged run (legacy CLI — transitional only)	56
15.4	3) Backend comparison and selection (legacy CLI)	57
15.5	4) Rollout comparison report (legacy CLI)	57
15.6	5) Standalone enricher run from mapper output	57
15.7	6) Standalone blind prediction	57
15.8	7) Manual rendering commands (this manual)	57
15.9	8) Distributed queue (legacy — one task)	58
15.10	9) Hyperparameter search grid (example)	58
15.11	Execution notes	58

16 Distributed MethyIValidation on shared storage	59
16.1 Purpose	59
16.2 Prerequisites	59
16.3 Layout (under <code>monte_carlo_runs/</code>)	59
16.4 Command sequence	60
16.5 Example paths (<code>/work</code> pattern)	60
16.6 Recovery	60
16.7 See also	61
17 Deployment and Distributed Workflow	62
17.1 Purpose	62
17.2 Architecture	62
17.3 Greenfield checklist	63
17.4 Database deploy (dual-backend)	63
17.4.1 PostgreSQL (recommended for new deployments)	63
17.4.2 Azure SQL (legacy / phase-1)	64
17.5 Action catalog and workflow definitions	64
17.6 Gateway configuration	65
17.7 Worker configuration	65
17.8 End-to-end operator flow	66
17.8.1 1. SamplePrepPipeline	66
17.8.2 2. Study validation / MC	66
17.8.3 3. Local workflow (no gateway)	66
17.9 Related documentation	67
18 Optional hyperparameter search	68
18.1 When to use	68
18.2 What it is	68
18.3 Command example	68
18.4 Pitfalls	69
18.5 See also	69
IV Part IV: Worked Tutorials	70
19 Tutorial: Healthy vs. Cancer (single group or increasing stages)	71
19.1 What you will build	71
19.2 Mental model (four layers)	71
19.3 Step 1 — Lay out cohort CSVs	71
19.4 Step 2 — Write the study manifest	72
19.4.1 Option A — Binary (healthy vs. one cancer group)	72
19.4.2 Option B — Increasing stages (healthy vs. stage 1 ... stage N)	72
19.5 Step 3 — Understand what <code>comparisons</code> does	73
19.6 Step 4 — Pick a profile	74
19.6.1 Quick picks (this tutorial)	74
19.6.2 Statistical modeling modes (<code>mc_*</code>)	74
19.6.3 Full profile catalog	75
19.7 Step 5 — Run it	76

19.8 Step 6 — Find the outputs	77
19.9 Recap	77
19.10See also	77
V Part V: Hyperparameter Versioning	78
20 Content-Addressed Action Store (CAAS)	79
20.1 Purpose	79
20.2 Default behavior	79
20.2.1 When to disable	79
20.3 Content key	79
20.4 On-disk layout	80
20.5 Hyperparameter set identity	80
20.6 Database mirror (distributed gateway)	80
20.7 Enabled actions	81
20.8 Skip, reuse, and force re-run	81
20.9 Observability checklist	81
20.10See also	81

1 MethylPipeline Usage Manual

1.1 What this manual is

This is the operational companion to the theory book. It is designed for execution teams who need:

- exact commands,
- required config keys,
- expected output artifacts,
- stage handoff checks,
- recovery actions when a run fails.

For statistical background, use [docs/theory/](#). For compiler and worker internals, use [docs/implementation/](#). Start with [Relationship to Theory and Implementation](#).

1.2 How to run staged validation

Use `methyl-workflow-run` with a DomainProgram and pipeline profile. That is the canonical path for new studies, local runs, and production gateway workers.

`methyl-validation --stability/--freeze/--model is legacy` — documented for transitional scripts and narrow recovery, not for new workflow design. See [Orchestration: workflow-run](#) before the stage chapters.

Goal	Start here
Understand entry points	Orchestration (ch.04)
First end-to-end run	Tutorial (ch.16)
One stage’s artifacts and gates	Part II (ch.05–09) — each chapter links back to workflow-run
Legacy CLI flags	<code>packages/methylvalidation/docs/USAGE.md</code>

1.3 Reading paths

- **Sample prep (FASTQ → HDF5):** Chapter 03 (`03-sample-prep-and-qc.qmd`).
- **New project setup:** Chapters 01, 02, 04 (orchestration), then 05 or [Tutorial ch.16](#).
- **Fast path (stage execution):** [Orchestration ch.04](#) → `methyl-workflow-run`; stage detail in ch.05–09.
- **Guided tutorial (healthy vs. cancer stages):** Chapter 16 — end-to-end walkthrough; §18.6 lists every shipped pipeline profile.

- **Hyperparameter versioning (CAAS):** 17-content-addressed-action-store.qmd (Part V; follows the tutorial).
- **Daily operations:** 10-artifacts-and-qa-checks.qmd through 12-command-cookbook.qmd.
- **Scaling MC across machines:** Chapter 13 — shared /work storage and queue subcommands.
- **Deployment (DB + gateway + workers):** Chapter 14.
- **Optional hyperparameter tuning:** Chapter 15 — methyl-hyperparam-search grid driver.
- **Standalone package usage:** Chapter 04.

1.4 End-to-end workflow map

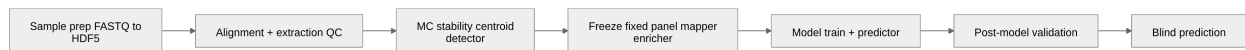


Figure 1.1: End-to-end pipeline stages

1.5 Scope boundaries

- **SamplePrepPipeline** (FASTQ → aligned BAM → methylation HDF5) runs before staged validation; see Chapter 3.
- **Canonical orchestration:** methyl-workflow-run + DomainProgram + profile — [chapter 04](#).
- **Legacy orchestration:** methyl-validation --stability/--freeze/--model — transitional scripts and recovery only; do not use for new studies.
- Blind-only prediction is documented as a standalone methyl-predictor operation.
- Package behavior is documented from implementation in packages/*.

1.6 Canonical references

- Sample prep workflow: [workflow_engine/sql_mssql/SamplePrepFlow.md](#)
- DomainProgram language: [docs/reference/domain-program-language.md](#)
- Pipeline stages diagram: [docs/architecture/pipeline-stages.md](#)
- Validation orchestration: [packages/methylvalidation/docs/USAGE.md](#)
- Distributed queue: [packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md](#)
- Hyperparameter search: [packages/methylvalidation/docs/HYPERPARAMETER_SEARCH.md](#)

2 Relationship to Theory and Implementation

This usage manual owns **operator commands**, stage handoffs, artifact gates, and deployment checklists. It does not derive statistical methods or document compiler internals.

Need	Read
DMP detection math, assumptions, limitations	Theory book — start with ch.03
DomainProgram language and schemas	Reference: domain-program-language
Parameter keys and defaults	Reference: configuration-reference
Workflow engine, workers, compiler	Implementation guide
System layers and orchestration paths	Architecture

Canonical orchestration: [methyl-workflow-run](#) with repo DomainPrograms, pipeline profiles, and `/work` study manifests. This is the path for new studies, local development, and production workers.

Legacy orchestration: `methyl-validation --stability/--freeze/--model` and the file-queue subcommands (`plan-runs`, `run-task`) remain documented for transitional scripts and recovery. Do not start new studies on the monolithic CLI — migrate to a DomainProgram that encodes the same stages.

Need	Read
Entry points, program selection	Orchestration (ch.04)
Path matrix (local / gateway / legacy)	Architecture: orchestration paths

Repo vs /work: profiles and programs live in the git repo; `project_*.json` and run artifacts live under `/work/<disease>/. See ch.02 and work-config-paths rule.`

Part I

Part I: Setup and Package Orientation

3 Prerequisites and Environment

3.1 Purpose

Set up a reproducible execution environment for all package CLIs and workflow stages.

3.2 Required tools

- Python environment in repo-local `.venv`
- Installed package CLIs (`methyl-*`)
- Quarto for manual rendering (optional for pipeline runs)
- TeX engine for PDF rendering (`quarto install tinytex` recommended)

3.3 Environment bootstrap

From repo root:

```
python3.12 -m venv .venv
source .venv/bin/activate
bash scripts/install_all.sh
```

Optional dependency sets:

```
bash scripts/install_all.sh --pipeline-reqs
bash scripts/install_all.sh --gpu-reqs
```

3.4 Verify CLI availability

```
source .venv/bin/activate
methyl-workflow-run --help
methyl-validation --help
methyl-centroid --help
methyl-detector --help
methyl-mapper --help
methyl-enricher --help
methyl-predictor --help
```

3.5 Runtime assumptions

- Activate `.venv` before every run.
- Use absolute paths in project configs where possible.
- Keep `samples_base_path` consistent across stages.

3.6 Success checks

- `.venv` activates without import errors.
- `methyl-workflow-run --help` and `methyl-validation --help` print options.
- Project JSON path resolves on the execution host.

3.7 Common failures and fixes

- **command not found:** activate `.venv` and reinstall with `scripts/install_all.sh`.
- **module import errors:** reinstall editable packages from repo root.
- **path mismatch between hosts:** set profile `actionConfig.validation.path_remap` or `--path-remap OLD=NEW`.

3.8 See also

- `README.md`
- `docs/DEPLOYMENT.md`

4 Project Config and Layout

4.1 Purpose

Define a **study manifest** (slim project JSON) plus reusable **profile**, **site**, and **program** artifacts so stability, freeze, model, and evaluation run without per-stage schema drift.

4.2 Four-layer configuration

Layer	Typical path	Owns
Study manifest	<code>/work/<disease>/configs/project_*.json</code>	Cohorts, stages, comparisons, paths, regulatory , validation_partitions , progression_order
Pipeline profile	<code>workflow_engine/domain/profiles/*.profile.json</code>	actionConfig tool parameters + scope booleans (<code>runDmpSelection</code> , ...)
Site manifest	<code>/work/site/methyl_site.json</code> (or <code>METHYL_SITE_CONFIG</code>)	Genomes, GTF, caches, cluster defaults
DomainProgram	<code>repo workflow_engine/domain/**/*.*.program.json</code>	Control flow, per-action with / stepOverride

Precedence (highest wins): program/instance override → profile **actionConfig** → analyte defaults (**regulatory.primary_analyte**) → site manifest → *(no Python fallback for tunable science knobs)*.

Study manifests **must not** contain `step_config`. Legacy projects: `python scripts/migrate_project_config.py --in-place path/to/project.json`.

4.3 Repository vs /work (what goes where)

Repository (versioned, documented)	<code>/work/<disease>/</code> (study-specific)
Pipeline profiles (<code>workflow_engine/domain/profiles/</code>)	<code>project_*.json</code> — cohorts, comparisons, sample paths, <code>output_base</code> , regulatory
DomainPrograms (<code>*.program.json</code> under <code>workflow_engine/domain/</code>)	Sample list CSVs under <code>data/</code>
JSON schemas (<code>schemas/config/</code>)	Run outputs (<code>monte_carlo_runs/</code> , <code>stability/</code> , ...)

Repository (versioned, documented)	/work/<disease>/ (study-specific)
Site example (tools/methyl-config-editor/configs/site_grch38.example.json)	/work/site/methyl_site.json (operator copy)

- Point `methyl-workflow-run --program` at a **repo** DomainProgram path (see [Orchestration](#)).
- Point `--context / projectPath` at a **study manifest on /work**.
- Pass `--context-file` with a profile from `workflow_engine/domain/profiles/`, or set `METHYL_PROFILE`.
- Set `METHYL_SITE_CONFIG` (or install site manifest under `/work/site/`).
- Do **not** treat repo `workflow_engine/domain/checks/*/configs/project_*.json` as production configs (CI smoke only).
- Do **not** copy `*.program.json` into `/work/<disease>/configs/`.

4.4 Minimum study manifest

Use one slim project file containing:

- global cohort metadata (controls, diseases, comparisons, optional stages[] / progression_order),
- filesystem anchors (output_base, samples_base_path),
- optional regulatory, validation_partitions, chromosomes, contexts.

Tool parameters live in the **profile** actionConfig sections (detection, validation, mapper, ...), not in the study manifest.

4.5 Validation / Monte Carlo keys (profile actionConfig.validation)

Resolved into `MonteCarloConfig` by `methyl-validation --project` (via `METHYL_PROFILE`):

- split policy: `train_fraction`, `n_iterations`, `seed`
- stability controls:
 - `run_stability`
 - `stability_dmp_freq`
 - `stability_min_balanced_accuracy`
 - `stability_gene_freq`
 - `stability_featurecuts_enabled`
 - `dmp_featurecuts_target_ba`, `gene_featurecuts_target_ba` (split BA gates)
 - `dmp_modeling_mode`, `gene_modeling_mode` (statistical axes)
 - `stability_gene_featurecuts_max_dmps`, `stability_gene_featurecuts_max_genes` (site/profile caps)
- freeze/model paths:
 - `freeze_stable_dmp_csv` (optional)
 - `production_output_dir` (optional)

- backend/model:
 - model_backend
 - feature_mode
 - ecdf_second_stage_enabled
 - backend-specific `tabular_*` / `generative_*` keys

Sample prep uses profile `actionConfig.alignment_qc` and `actionConfig.extraction_qc` ([Chapter 3](#)).

4.6 Output layout

Most runs write to:

`<output_base>/<project_name>/monte_carlo_runs/`

Important subtrees:

- `run_XXXX/` (per-iteration artifacts)
- `stability/` (stable panel and summaries)
- `production/` (freeze and final model outputs)
- `model_mc/` (backend comparison runs)
- `post_model_validation/` (frozen-model descriptive evaluation)

4.7 Handoff model

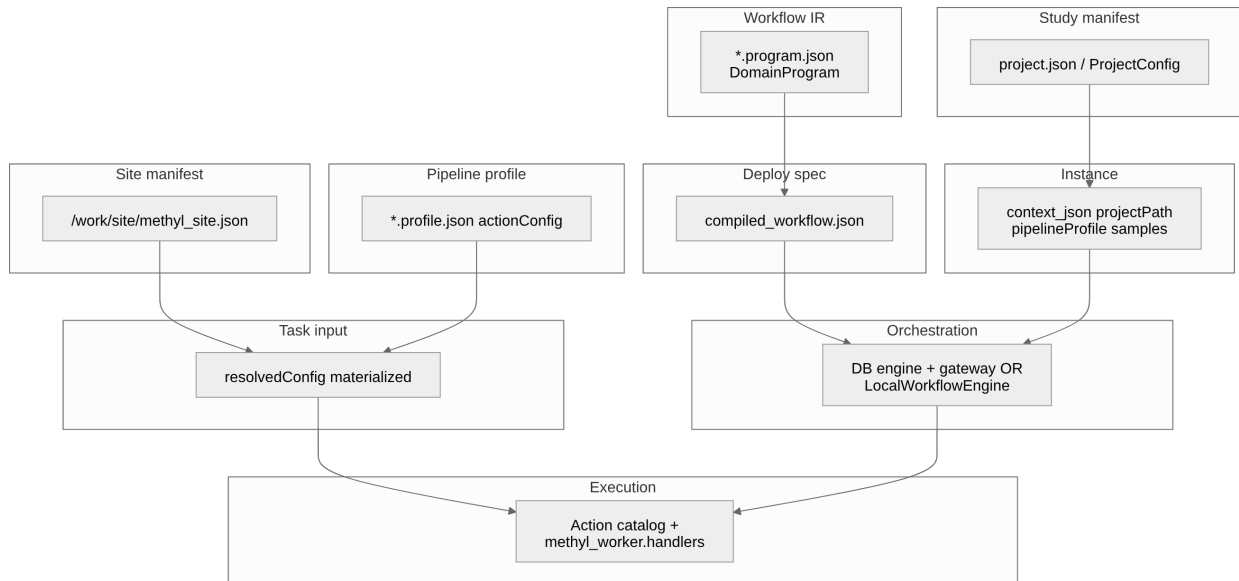


Figure 4.1: Configuration layer model

4.8 Do not do this

- Do not embed tool parameters in the study manifest (`step_config` is rejected at load).
- Do not put blind-only predictor config into validation-driven runs.
- Do not change `samples_base_path` between stages without explicit remap.
- Do not run `--freeze` before stable panel exists.

4.9 Success checks

- Study manifest loads via `methy1-utils / methy1-validation --project ....`
- `METHYL_PROFILE` resolves `actionConfig.validation` (or pass `--config mc.json`).
- Run directories appear under expected root.

4.10 See also

- `docs/reference/config-parameter-matrix.md`
- `docs/reference/domain-program-language.md`
- `docs/theory/chapters/11-project-configuration.qmd`
- `docs/reference/configuration-reference.qmd`

5 Sample Prep and Quality Control

5.1 Purpose

SamplePrepPipeline is the per-sample upstream workflow that turns raw WGBS FASTQs into methylation HDF5 files (`{chromosome}-CG.h5`) that downstream stages consume. Every sample passes through **two automated guardrails** before it enters stability, freeze, or discovery workflows:

1. **Alignment QC** (`methyl-qc`) — confirms Parabricks alignment and sequencing metrics are acceptable (or fixable via trim/realign).
2. **Extraction QC** (`methyl-extraction-qc`) — confirms MethylExtractor produced sufficient coverage and conversion-quality signal across chromosomes.

Without these gates, bad alignments would waste GPU extraction time, and under-covered or poorly converted extractions would pollute centroids, DMP discovery, and classifiers.

Each sample is aligned on GPU using either **linear** (`fq2bam_meth`) or **pangenome** (`vg giraffe` via Clara Parabricks), then passes through the same QC and extraction gates below.

Workflow source of truth: [workflow_engine/domain/fixtures/sample_prep.program.json](#)

Operator reference: [workflow_engine/sql_mssql/SamplePrepFlow.md](#)

Implementation detail (alignment modes, storage, guardrails): [docs/implementation/sample-preparation-flow.md](#)

5.2 Alignment modes (linear vs pangenome)

After FASTQ download, SamplePrep chooses **one GPU alignment path** per sample (and the same path on fastp remediation realign):

Table 5.1: Alignment modes

Mode	Scope flag	Worker action	Parabricks entry	Reference
Linear (default)	<code>usePangenome: false</code>	<code>sample.parabricks_fq2bam</code>	<code>fq2bam_meth</code> — traditional bisulfite-aware WGBS align to a linear reference	<code>referenceFasta</code> on instance / site genomes

Mode	Scope flag	Worker action	Parabricks entry	Reference
Pangenome	<code>usePangenome:</code> <code>true</code>	<code>sample.parabricks_giraffe</code>	<code>giraffe</code> — vg Giraffe on an HPRC graph bundle, surjected to GRCh38 coordinates; then <code>collectmultiplemetrics</code> for the same QC tar/JSON shape as linear	Site manifest <code>pangenome_genome</code> (<code>gbz</code> , <code>dist</code> , <code>min</code> , <code>zipcodes</code> , <code>ref_paths</code> , <code>linear_ref_fasta</code>)

How to enable pangenome: set instance `alignmentMode`: "pangenome" (binds `usePangenome`) or profile `actionConfig.parabricks.alignment_mode`: "pangenome". Default is **linear**.

WGBS note: stock HPRC graphs are not bisulfite-aware. For pangenome WGBS, operators must stage a **C→T-converted / WGBS-compatible graph** (or document an explicit read-conversion workflow at the site). See implementation guide § Parabricks pangenome alignment and `scripts/download_pangenome_hprc_grch38.sh`.

Downstream steps (`methy1_qc`, `fragmentomics`, `methy1_extract`, extraction QC) are identical for both modes — alignment QC consumes the same normalized metrics JSON regardless of aligner.

5.3 End-to-end flow

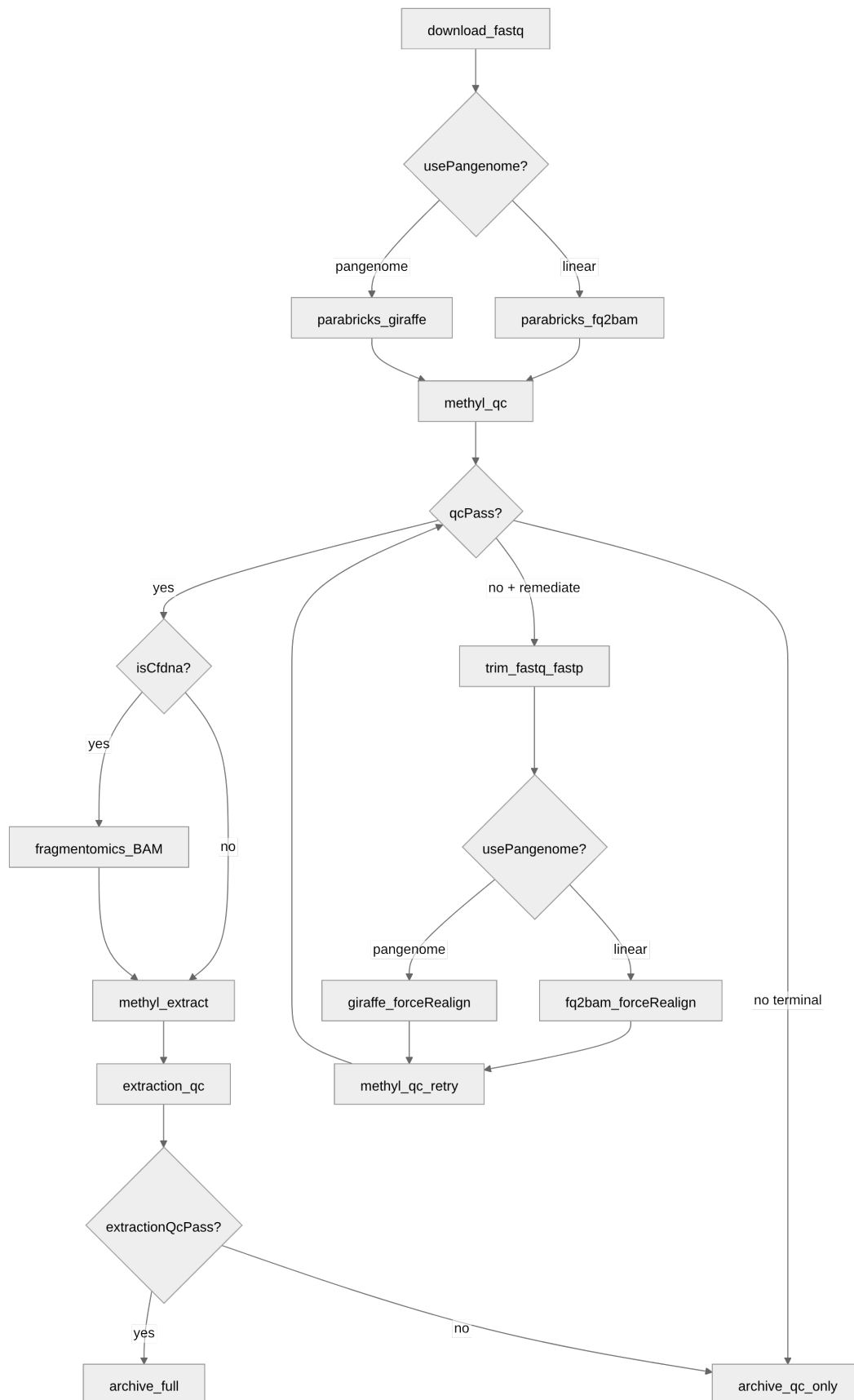


Figure 5.1: Sample prep and QC flow (linear or pangenome align, then shared QC gates)

Gate	Scope variable	When	Blocks
Alignment	qcPass	After Parabricks, before extract	Extraction (unless remediation succeeds)
Extraction	extractionQcPass	After MethylExtractor	Full archive and BAM deletion

Ordering rationale: alignment QC runs before cfDNA fragmentomics so failed samples skip BAM scanning. Fragmentomics and extraction both need the aligned BAM. FASTQs are retained until QC is fully resolved so **fastp** can trim and Parabricks can realign on the remediation path.

5.4 Where artifacts live

Per sample under `/work/samples/{sample_id}/`:

Artifact	Role
<code>*deduplicate_metrics.txt</code>	Picard-style duplication metrics from Parabricks
<code>{sample_id}.json</code>	Parabricks WGBS metrics JSON (sequencing + alignment)
<code>{sample_id}.qc-metrics.tar</code>	Optional tar of tabular metrics (used when JSON is absent)
<code>{output_base}/{project}/alignment_qc/{sample_id}.json</code>	Normalized V2 alignment QC export (when run project-scoped)
<code>{sample_id}.extraction_manifest.json</code>	MethylExtractor summary (written at extract time)
<code>{sample_id}.extraction_qc.json</code>	Extraction guardrail report
<code>{sample_id}.sample_prep_log.jsonl</code>	Append-only audit of every prep action
<code>{chr}-CG.h5</code>	Methylation matrices consumed by methyl-centroid

5.5 Alignment QC: inputs and normalized JSON

methyl-qc (packages/methylalignmentqc) combines **two upstream metric families** into one decision:

5.5.1 Picard-style duplication metrics

Parabricks emits a tab-separated **deduplication metrics** file (names vary: `*deduplicate_metrics.txt` or `*duplication_metrics.txt`). The parser (`methyl_alignment_qc/core/parser.py`) reads Picard-compatible columns such as:

- LIBRARY, UNPAIRED_READS_EXAMINED, READ_PAIRS_EXAMINED
- SECONDARY_OR_SUPPLEMENTARY_RDS, UNMAPPED_READS
- UNPAIRED_READ_DUPLICATES, READ_PAIR_DUPLICATES
- PERCENT_DUPLICATION

These feed `summary_stats` on the export JSON (total reads, duplication rate, etc.). Optional config guardrails (`actionConfig.alignment_qc.optional_guardrails.duplication_rate_max`) can fail samples with excessive PCR duplication even when cycle quality looks fine.

5.5.2 Parabricks WGBS metrics JSON

Each sample directory should contain `{sample_id}.json`: Parabricks’ structured report with sections such as:

- `quality_yield` — total/PF reads and Q30 bases
- `mean_quality_by_cycle` — per-cycle mean Phred (Read 1 then Read 2; for 151+151 cycles, Read 2 starts at cycle 152)
- `gc_bias_summary`, `insert_size_metrics`
- `pre_adapter_summaries` — deamination and OxoG artifact qscores

When this file is present, guardrails in `wgbs_parabricks_qc.py` evaluate sequencing and alignment health **before** methylation extraction.

5.5.3 Our equivalent export JSON (V2)

The worker writes a normalized **V2 row-oriented** JSON per sample (schema: `schemas/config/alignment_qc/exp`). Key blocks:

Block	Contents
<code>summary_stats</code>	Duplication rate, read counts from Picard metrics
<code>quality_yield</code> , <code>mean_quality_by_cycle</code> , ...	Parabricks sections (embedded or referenced)
<code>guardrails.details</code>	Per-metric pass/fail, observed value, normal range, operator message
<code>guardrails.screening</code>	Cycle-quality disposition and recommended trim counts
<code>guardrails.overall_pass</code>	Single boolean bound to workflow <code>qcPass</code>
<code>qc_history</code>	Append-only list of evaluations (initial + post-remediation retries)
<code>fragmentomics_metrics</code>	cfDNA insert-size guardrails when enabled

V1-shaped payloads are assembled internally and converted to V2 for export; standalone tools can migrate older files with `methy1-qc-migrate-guardrails`.

5.6 Alignment guardrails: what “good alignment” means

Core WGBS guardrails (defaults in `wgbs_parabricks_qc.py`) include:

Metric	Default expectation	Why it matters
PF read percent	90%	Run-level filtering quality
Q30 percent	85% (configurable)	Base-call confidence
Mean quality	35	Overall sequencing noise
Min quality after cycle 20	30	Late-cycle degradation

Metric	Default expectation	Why it matters
AT / GC dropout	< 3 / < 5	Coverage bias across genome
Median insert size	150–300 bp	Library fragmentation sanity
Deamination qscore	30	Expected WGBS conversion signal
OxoG qscore	20	Oxidative damage risk

Each metric carries a **message** explaining the biological/operational concern. `guardrails.overall_pass` is the logical AND of all evaluated checks (including optional and analyte-specific add-ons).

5.6.1 Optional and analyte-specific checks

Configure under profile or site `actionConfig.alignment_qc` (not in the study manifest):

```
{
  "alignment_qc": {
    "optional_guardrails": {
      "duplication_rate_max": 0.25,
      "min_pf_reads": 1000000
    },
    "bisulfite_conversion": {
      "enabled": true,
      "source": "auto",
      "min_conversion_rate_pct": 99.0
    },
    "fragmentomics": {
      "enabled": true,
      "profile": "cfdna"
    },
    "cycle_screening": {
      "enabled": true,
      "r2_quality_threshold": 30,
      "max_trim_bases": 8
    },
    "alignment_guardrails": {
      "enabled": true,
      "min_mapping_rate": 0.98,
      "flagstat_enabled": true,
      "min_properly_paired_rate": 0.90
    }
  }
}
```

- **Alignment guardrails:** mapping rate, secondary/supplementary rate, GC uniformity (Picard-derived), and properly paired rate (`samtools flagstat` on the BAM). Enabled by default for `cfdna` and `buffy_coat` via analyte profile. See [implementation guide](#).

- **Bisulfite conversion:** reads `bisulfite_conversion.json` sidecar when present; otherwise may use deamination qscore as a qualitative proxy.
- **cfDNA fragmentomics:** when `primary_analyte` is `cfdna`, insert-size histogram guardrails can flag abnormal fragment profiles before extraction.

See [docs/ANALYTE_PROFILES.md](#) for profile defaults.

5.7 Cycle screening and automated recovery

Historically, a single `min(mq[20:])` over the combined cycle series could fail fixable Read-2 start dips with a generic “do not proceed” message. **Cycle screening** (`cycle_quality_screening.py`) inspects Read 1 and Read 2 separately and assigns a **disposition**:

Disposition	Meaning	SamplePrep action
USE_CURRENT_ALIGNMENT	Guardrails pass	Continue to extract
REALIGN_TRIM / REALIGN_READ2_TRIM	Fixable read-end quality dip	<code>sample.trim_fastq</code> → Parabricks <code>forceRealign</code> → <code>methyl-qc</code> retry
INVESTIGATE_MULTI_REGION	Multiple low-quality regions	Manual review; no auto-trim
INVESTIGATE_GUARDRAIL_ONLY	Cycles OK; duplication/depth/insert fail	Manual review
NOT_FIXABLE	No automated path	Archive QC-only artifacts, mark failed

Screening writes `guardrails.screening` with recommended trim bases (`trim_front1`, `trim_tail1`, `trim_front2`, `trim_tail2`). SamplePrep binds `remediateAlignment` when disposition is trim-eligible and FASTQs are still on disk.

Important: FASTQs must **not** be deleted before final alignment QC disposition. The workflow deletes FASTQs only after pass or final reject (including post-remediation retry).

Offline cohort review (no re-align):

```
source .venv/bin/activate
python scripts/alignment_qc_cohort_screening.py \
  --qc-dir /work/projects/prostate-cancer/alignment_qc \
  --group healthy=/work/projects/prostate-cancer/data/healthy.csv \
  --group pca=/work/projects/prostate-cancer/data/pca.csv \
  --out /work/projects/prostate-cancer/alignment_qc/screening_report
```

5.8 Extraction QC: confirming enough good reads

Alignment QC validates the **BAM**. Extraction QC validates the **methylation call layer** MethylExtractor produced.

5.8.1 Input: extraction manifest

MethylExtractor writes `{sample_id}.extraction_manifest.json` with:

- `summary` — genome-wide CpG weighted mean coverage, CHH/CHG methylation levels
- `per_chromosome` — per-chromosome blocks with `CG.mean_coverage` and related stats

5.8.2 Output: extraction QC JSON

`methyl-extraction-qc` reads the manifest and writes `{sample_id}.extraction_qc.json`. Default guardrails (`methylextractionqc/guardrails.py`):

Guardrail	Default	Interpretation
<code>cpg_weighted_mean_coverage</code>	10×	Enough reads at CpG sites passing minimum coverage
<code>chh_methylation_level</code>	0.02	Low non-CpG methylation (conversion sanity) when CHH extracted
<code>chg_methylation_level</code>	0.02	Same for CHG
<code>chromosome_completeness</code>	all expected chromosomes	Every project chromosome present with CG metrics
<code>chromosome_uniformity</code>	autosomal min/median	0.5 No severe localized dropout
<code>read_discard_fraction</code>	0.9	Catastrophic read loss (<code>1 - read_retention_rate</code>) from extraction-time filters

`guardrails.overall_pass` maps to workflow `extractionQcPass`. Failures are **terminal** today (no automatic retry loop).

Override thresholds via profile `actionConfig.extraction_qc`. Schema: `schemas/extraction_qc_output.schema`

5.8.3 Why both gates matter

Question	Answered by
Did sequencing and alignment produce a usable BAM?	Alignment QC
Did we extract enough confident methylation signal genome-wide?	Extraction QC

A sample can pass alignment but fail extraction (e.g., coverage collapse on certain chromosomes). Conversely, remediation after alignment QC prevents spending extract GPU time on BAMs that will never support stable downstream statistics.

5.9 Pass, fail, and handoff

Outcome	Alignment QC	Extraction QC	Next step
Pass both	<code>qcPass: true</code>	<code>extractionQcPass: true</code>	Optional full archive → delete FASTQs/BAM → sample ready for DataDrivenPipeline
Fail alignment (not fixable)	<code>qcPass: false</code>	—	QC-only archive, mark failed, skip extract
Fail after remediation	<code>qcPass: false</code> (attempt 2)	—	Same as above
Pass alignment, fail extraction	<code>qcPass: true</code>	<code>extractionQcPass: false</code>	QC-only archive, mark failed; H5 may exist but must not enter production cohorts

When SamplePrep instance **COMPLETED**, start stability or discovery workflows with the same `projectPath`. Downstream code resolves HDF5 via `methylation_domain.helpers.resolve_methylation_h5_path()` — do not hard-code sample paths.

Workers tag scope with **MethylSampleRef** including `alignmentQc`, `extractionQc`, and `methylation` refs for traceability.

5.10 Configuration checklist

Before starting SamplePrep, confirm:

- ☐ `project.json` lists samples and `samples_base_path` / `path_remap` resolve to `/work/samples/...`
- ☐ Profile/site `actionConfig.alignment_qc` thresholds match analyte (cfDNA vs buffy coat)
- ☐ Profile `actionConfig.extraction_qc` min coverage appropriate for WGBS depth expectations
- ☐ `validation.regulatory.primary_analyte` set (drives fragmentomics profile)
- ☐ Instance `context_json` includes `fastqStorage`, `referenceFasta`, `samples[]`
- ☐ When using pangenome: site `pangenome_genome` bundle staged, `alignmentMode: "pangenome"` (or profile `parabricks.alignment_mode`)

5.11 Commands (standalone debugging)

```
source .venv/bin/activate

# Alignment QC for one sample directory
methylation-qc --samples /work/samples/SAMPLE_ID --output-dir /work/projects/prostate-cancer/alignment

# Project-scoped (uses profile actionConfig.alignment_qc via METHYL_PROFILE)
methylation-qc --project /work/projects/prostate-cancer/configs/project_Example.json

# Extraction QC after MethylExtractor
methylation-extraction-qc --sample-dir /work/samples/SAMPLE_ID --sample-id SAMPLE_ID
```

5.12 Related documentation

- Implementation guide (guardrail boundaries, remediation, extraction filtering, cfDNA fragmentomics): [docs/implementation/sample-preparation-flow.md](#)
- Workflow operator guide: [workflow_engine/sql_mssql/SamplePrepFlow.md](#)
- Architecture §0: [workflow_engine/docs/pipeline_architecture.md](#)
- Worker contract: [workflow_engine/contract/sample_prep_capabilities.md](#)
- Alignment QC package: [packages/methylalignmentqc/docs/USAGE.md](#)
- Extraction QC package: [packages/methylextractionqc/README.md](#)
- Historical design notes (cycle screening): [docs/plans/alignment-qc-screening.plan.md](#)

6 Orchestration: methyl-workflow-run (canonical)

6.1 Purpose

Choose the right **entry point** for staged validation. New studies and production runs should use **methyl-workflow-run** with a DomainProgram and pipeline profile. The monolithic **methyl-validation --stability/--freeze/--model** path is **legacy** — kept for transitional scripts and one-off recovery, not for new workflow design.

6.2 Preferred vs legacy

Canonical (use this)	Legacy (transitional only)
Entry point Topology DomainProgram JSON (workflow_engine/domain/**/.program.json) Parameters Profile actionConfig + site manifest + optional instance overrides	methyl-validation --stability, --freeze, --model, ... Hard-coded stage sequence inside the CLI Same keys, but resolved only through project path and CLI flags
Resume Per-action .action_results/ manifests in the / compiled graph	--resume, --skip-detection, coarse stage flags
Production cluster methyl-gateway + methyl-worker (same programs as local)	plan-runs / run-task file queue (legacy distributed MC)
When to use All new studies, CI smoke, portal/gateway instances	Existing shell scripts, queue workers not yet migrated, narrow recovery

See also [Architecture: orchestration paths](#).

6.3 What you pass

Every canonical run needs three inputs:

1. **--program** — workflow topology (which actions run, in what order, FOREACH loops, gates).
2. **--context-file** — pipeline profile (`*.profile.json`) with merged `actionConfig`.
3. **--context** — JSON with at least `projectPath` (study manifest on `/work`) and usually `pipelineProfile`.

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/buffy_healthy_vs_pca/configs/buffy_mc_stability.program \
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_Pca/mc_dmp_gene_fc.profile.json"}' \
  --parallel-workers 1
```

In production, point `--program` and `--context-file` at the promoted runtime bundle, e.g. `/work/epimethyl/current/runtime-bundle/domain/...` (see [Deployment and distributed workflow](#)).

6.4 Programs by scope

Pick a DomainProgram that matches how much of the lifecycle you need. Stage chapters (ch.05–09) describe artifacts and gates; this table maps them to programs.

Scope	Example program	Covers
MC stability only	<code>buffy_mc_stability.program.json</code> , <code>pca1_5_mc_stability.program.json</code>	Centroid + detector loops, stability, freeze-readiness gate
Freeze only (after stability)	<code>pca1_5_freeze.program.json</code>	<code>prepare_freeze_project</code> , mapper, enricher, progression
Full study lifecycle	<code>study_validation_lifecycle.program.json</code>	Stability → freeze → model MC → selection → hold-out

Binary healthy-vs-cancer walkthrough: [Tutorial: healthy vs. cancer stages.](#)

6.5 Local development without a database

```
methyl-workflow-run \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/pca1_5_mc_stability_smoke.program.
  --context-file workflow_engine/domain/profiles/mc_dmp.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-
  --stub-external
```

Validate compile/bindings before a long run: `python workflow_engine/domain/checks/<study>/check_pipeline.py --program ....`

6.6 Legacy methyl-validation (when it still appears)

Use `methyl-validation` only when:

- You must run an existing shell script or notebook that has not been migrated.
- You need a **narrow recovery** command (e.g. `--stability --skip-detection` after MC iterations exist).
- You operate the **file-queue** distributed path (`plan-runs` → `run-task` → `aggregate-results`) — see [Distributed MethylValidation](#).

Do **not** start new studies with `methyl-validation --stability` as the primary entry. Migrate to a `DomainProgram` + profile (same `actionConfig` keys; different invocation).

```
# Legacy example - prefer methyl-workflow-run for new work
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

Package-level detail and CLI flag reference: `packages/methylvalidation/docs/USAGE.md` (documents legacy flags for parity with workflow actions).

6.7 Related

- [Project config and layout](#) — four-layer config model
- [DomainProgram language](#) — program vs profile vs manifest
- [Command cookbook](#) — copy-ready sequences (canonical first)
- [Architecture: orchestration paths](#)

7 Package Reference (Individual Usage)

7.1 Purpose

Summarize each active package as a standalone tool, then map it to staged `methylation-validation` orchestration.

For the full SamplePrep story (Picard metrics, Parabricks JSON, guardrails, remediation, extraction QC), see [Chapter 3 — Sample Prep and Quality Control](#).

7.2 Package quick map

Package	CLI	Primary role	Typical standalone output
<code>methylationalignmentqc</code>	<code>methylation-qc</code>	Parse alignment/dup metrics into normalized JSON	<code>alignment_qc/<sample>.json</code>
<code>methylationcentroid</code>	<code>methylation-centroid</code>	Build cohort centroids from methylation files	centroid artifacts per group
<code>methylationdetector</code>	<code>methylation-detector</code>	Discover/classify DMPs	<code>dmps-**-discovery.csv</code> , <code>result*.json</code>
<code>methylationmapper</code>	<code>methylation-mapper</code>	Map DMP loci to genes/features	<code>all-gene_name-combined.csv</code>
<code>methylationenricher</code>	<code>methylation-enricher</code>	Enrichment/module interpretation	<code>enrichment_merged.csv</code> , <code>modules_ranked.csv</code>
<code>methylationclassifier</code>	<code>methylation-classifier</code>	Train production classifier (ECDF path)	<code>classifiers/*.pkl</code>
<code>methylationpredictor</code>	<code>methylation-predictor</code>	Predict and report metrics/inference	<code>predictors/*/validation_metrics.json</code> or blind reports
<code>methylationvalidation</code>	<code>methylation-validation</code>	Orchestrate MC/stability/freeze/model workflows	<code>monte_carlo_runs/*</code>
<code>methylationdiseaseprogression</code>	<code>methylation-disease-progression</code>	Cross-stage synthesis from mapper/enricher outputs	<code>progression/*.csv</code> , <code>summary.json</code>

7.3 Standalone examples

7.3.1 Alignment QC

```
source .venv/bin/activate
methyl-qc --metrics-root /data/metrics --output-dir /work/alignment_qc
```

7.3.2 Enrichment from mapper output

```
source .venv/bin/activate
methyl-enricher --input /work/project/mapper/pca_pca1/all-gene_name-combined.csv --modules --o
```

7.3.3 Blind prediction with frozen model

```
source .venv/bin/activate
methyl-predictor --project /work/project/monte_carlo_runs/production/project.json
```

7.4 Orchestrated equivalents in methyl-validation

- Stability stage runs centroid+detector per iteration.
- Freeze stage runs centroid->detector->mapper->enricher (and optional progression).
- Model stage runs:
 - `ecdf`: classifier->predictor
 - `tabular_sklearn` / `generative_hybrid`: in-process backend train/predict
- Post-model validation runs frozen-model descriptive holdouts.

7.5 Integration diagram

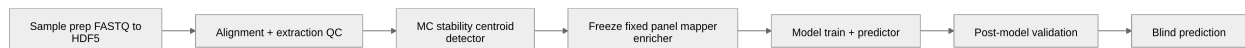


Figure 7.1: Package integration

7.6 Do not do this

- Do not assume standalone package defaults match project-orchestrated overrides.
- Do not skip `project.json` artifact handoff checks when chaining packages manually.

7.7 See also

- `README.md`
- `packages/*/docs/USAGE.md`

Part II

Part II: Staged Solution Workflow

8 Stage: Stability

8.1 Purpose

Run repeated train/validation iterations to identify DMPs that recur across runs and materialize a stable production panel.

Canonical entry: [Orchestration \(workflow-run\)](#). Use `methyl-workflow-run` with an MC stability DomainProgram. The `methyl-validation` commands in this chapter are **legacy**.

8.2 Entry command (canonical)

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/buffy_healthy_vs_pca/configs/buffy_mc_stability.prog
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_PC
  --parallel-workers 1
```

8.3 Legacy entry command

For transitional scripts only — prefer workflow-run for new studies:

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

Optional strict FeatureCuts alignment (legacy CLI):

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
  --stability --stability-featurecuts --stability-target-ba 0.95 --stability-min-selected-dmps
```

8.4 Required config keys

In the pipeline profile `actionConfig.validation` (set `METHYL_PROFILE` or pass `--context-file`):

- `n_iterations`, `train_fraction`, `seed`
- `run_stability`
- `stability_dmp_freq`
- `stability_min_balanced_accuracy`
- `stability_gene_freq`
- optional adaptive stop controls:
 - `stability_early_stop_enabled`
 - `stability_min_iterations`
 - `stability_convergence_window`
 - `stability_convergence_jaccard`
 - `stability_convergence_max_size_delta`
 - `stability_convergence_patience`
- optional FeatureCuts controls:
 - `stability_featurecuts_enabled`
 - `dmp_featurecuts_target_ba`, `gene_featurecuts_target_ba` (split BA gates)
 - `dmp_modeling_mode`, `gene_modeling_mode`
 - `stability_gene_featurecuts_max_dmps`, `stability_gene_featurecuts_max_genes` (site/profile caps)

8.5 Statistical modeling modes (generic profiles)

Profiles are **process-agnostic**; study facts stay in the study manifest. See `docs/reference/domain-program-language.md` for the full matrix.

Goal	Profile	Key modes
DMP recurrence only	<code>mc_dmp</code>	<code>dmp_modeling_mode: raw_pool</code>
BA-gated DMP panel	<code>mc_dmp_fc</code>	<code>dmp_modeling_mode: featurecuts</code>
Enricher gene recurrence (PPI or Enrichr)	<code>mc_gene</code>	<code>stability_gene_recurrence_source: enricher</code> , <code>enricher.ppi_only</code>
Gene FeatureCuts on enricher/PPI genes	<code>mc_gene_fc</code>	<code>gene_modeling_mode: featurecuts</code>
Full DMP + gene FC (production)	<code>mc_dmp_gene_fc</code>	both axes featurecuts
Two-phase panel lock	<code>phase_a_dmp_stability</code> → <code>phase_b_gene_from_stable_dmps</code>	stable DMP panel then gene FC

Canonical invocation (same program/profile as above):

```

methyl-workflow-run \
  --program workflow_engine/domain/checks/buffy_healthy_vs_pca/configs/buffy_mc_stability.prog
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_P

```

Mapper CSV pattern follows modeling mode: discovery → **dmps-*-discovery.csv**; featurecuts → **dmps-*-selected.csv**. Downstream consumers should prefer **dmps-*-selected.csv**.

Deprecated names (`mc_dmp_discovery`, `mc_dmp_featurecuts`, `mc_gene_mapper`, `mc_gene_featurecuts`, `discovery_gene_featurecuts`, `dmp_panel_stability`, `gene_enricher_stability`, `buffy_mc_gene_fc`) alias to the canonical `mc_*` presets in `pipeline_profiles.py`.

8.6 Process model

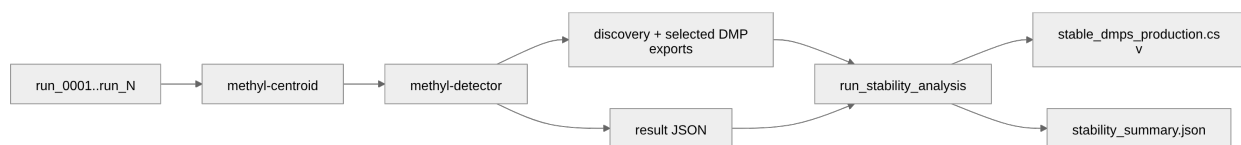


Figure 8.1: Stability stage process model

8.7 Expected outputs

- `<mc_root>/stability/stable_dmps_production.csv`
- `<mc_root>/stability/stability_summary.json`
- `<mc_root>/all_metrics.csv`
- `<mc_root>/metrics_summary.json`
- `<mc_root>/baseline_manifest.json`

Where `<mc_root>` is: `<output_base>/<project_name>/monte_carlo_runs/`

8.8 Success checks

- `stable_dmps_production.csv` exists and is non-empty.
- `all_metrics.csv` has successful rows near expected run count.
- `stability_summary.json` reports configured thresholds.
- if adaptive stop is enabled, `stability_summary.json` includes `early_stopping` diagnostics (triggered, checkpoint history, and stop iteration when applicable).

8.9 Do not do this

- Do not proceed to freeze when stable panel is missing or empty.
- Do not interpret blind outputs as stability quality evidence.

8.10 Recovery

- **Re-run workflow after MC complete (recommended):** re-run the same DomainProgram. Each action inside the MC FOREACH loop skips independently when its `.action_results/` manifest matches; post-MC steps (stability, readiness) skip the same way once they have run successfully once.

```
source .venv/bin/activate
methyl-workflow-run --program workflow_engine/domain/checks/.../buffy_mc_stability.program.json
--context-file workflow_engine/domain/profiles/your.profile.json
```

- **Stability-only (CLI, no workflow):** when MC outputs exist but stability never succeeded:

```
methyl-validation --project /work/.../configs/project.json --stability --skip-detection
```

- Resume interrupted MC iteration loop:

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

- If panel is too small: lower `stability_dmp_freq` or increase `n_iterations`.
- If run quality is too low: tune detector and/or adjust `stability_min_balanced_accuracy`.

8.11 See also

- docs/theory/chapters/12-two-workflows.qmd
- docs/theory/chapters/15-model-creation-and-validation.qmd
- If many MC iterations are split across several hosts, use shared `output_base` and the distributed queue flow: [Distributed MethylValidation on shared storage](#) (Chapter 13).

9 Stage: Freeze

9.1 Purpose

Create frozen production artifacts using the stable panel and generate biological interpretation outputs (mapper/enricher, optional progression).

Canonical entry: [Orchestration \(workflow-run\)](#). After stability, run a freeze Domain-Program (e.g. `pca1_5_freeze.program.json`) or the full `study_validation_lifecycle.program.json`. Legacy `methyl-validation --freeze` is transitional only.

9.2 Entry command (canonical)

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/pca1_5_freeze.program.json \
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-C
```

9.3 Legacy entry command

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

9.4 Prerequisites

- Stability stage completed.
- Stable panel available at:
 - configured profile `actionConfig.validation.freeze_stable_dmp_csv`, or
 - default `<mc_root>/stability/stable_dmps_production.csv`.

9.5 Process model

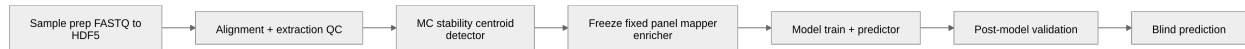


Figure 9.1: Freeze stage in pipeline lifecycle

9.6 Expected outputs

- `<mc_root>/production/project.json`
- `<mc_root>/production/production_summary.json`
- `<mc_root>/production/model_bundle/mapper_dmp_annotations.csv` (when mapped-family feature bundles are prepared)
- mapper outputs under `production/mapper/...`
- enricher outputs under `production/enricher/...`
- optional progression outputs under `production/progression/...`

9.7 Success checks

- `production/project.json` exists and includes fixed panel usage for detector.
- expected mapper and enricher files are present.
- stage-specific logs show successful return codes.
- for non-dmp observed-hybrid families, `production_summary.json` includes `mapper_annotation_cache` metadata.

9.8 Recommended post-freeze gate

Run biological readiness before model training:

```
source .venv/bin/activate
methyl-validation biological-readiness /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG
```

This chained gate verifies:

- enricher completeness (`verify-complete`),
- strict progression input completeness (`--strict-missing`),
- stability/freeze/progression readiness verdict.

To include advisory Grok commentary in the readiness section:

```
methyl-validation biological-readiness /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG --g
```

Without `--grok-review`, the chain keeps Grok disabled by default and still enforces deterministic checks.

9.9 Do not do this

- Do not run freeze before stability panel exists.
- Do not delete `production/project.json`; later model and predictor stages depend on it.

9.10 Recovery

- If freeze fails because stable panel missing, rerun stability or set `freeze_stable_dmp_csv`.
- If path mismatch occurs across machines, use `path_remap` or `--path-remap`.

9.11 See also

- `packages/methylvalidation/docs/USAGE.md`
- `docs/usage/index.qmd`

10 Stage: Model

10.1 Purpose

Train and select final production model artifacts after freeze and biological-readiness checks.

Canonical entry: [Orchestration \(workflow-run\)](#). Model training and selection are steps in `study_validation_lifecycle.program.json` (or portal/gateway instances compiled from the same programs). Legacy `methyl-validation --model / --model-mc` flags are transitional only.

10.2 Common execution patterns

10.2.1 Pattern A: Full lifecycle (canonical)

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/study_validation_lifecycle.program
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-CG'}
```

Run biological readiness before model stages when not using the full lifecycle program:

```
methyl-validation biological-readiness /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG
```

10.2.2 Pattern B: Direct production model build (legacy CLI)

```
source .venv/bin/activate
methyl-validation biological-readiness /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG
```

10.2.3 Pattern C: Backend model-MC comparison then select best (legacy CLI)

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
--select-best-model --model-mc-all --selection-metric balanced_accuracy --selection-stat med.
```

10.3 Backend behavior

- `ecdf`: `methyl-classifier` -> `methyl-predictor`
- `tabular_sklern`: in-process bundle -> train -> predict
- `generative_hybrid`: in-process bundle -> train -> predict

For `ecdf` + observed-hybrid mapped families (`feature_mode=observed_hybrid`, `feature_family_set!=dmp_score`) the runtime path is aggregated ECDF OvR package train/predict and skips ECDF second-stage refinement.

Configured by profile `actionConfig.validation.backend_profiles` (plus optional CLI backend override).

10.4 Process model

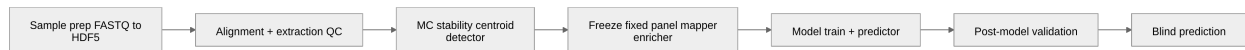


Figure 10.1: Model stage in pipeline lifecycle

10.5 Required config keys

- `backend_profiles`
- backend-specific `params` under each profile (for example `feature_mode`, `ecdf_second_stage_enabled`, `tabular_methods`, `generative_latent_dim`)
- observed-hybrid family contract (when used):
 - `feature_family_set`
 - mapper annotation cache from freeze (`production/model_bundle/mapper_dmp_annotations.csv`)
- tabular controls:
 - `tabular_methods`
 - `tabular_method_selection_metric`
 - optional `tabular_max_dmps` (null/0 keeps all stable loci)
- generative controls:
 - `generative_latent_dim`, `generative_epochs`, and related keys
- optional gate:

- require_biological_review_for_model
- biological_review_confirmed

Legacy flat backend keys are rejected. Migrate older configs with:

```
methvl-validation-migrate-backend-config /path/to/project.json --in-place
```

10.6 Expected outputs

- <mc_root>/production/classifiers/...
- <mc_root>/production/predictors/...
- aggregated ECDF mode writes:
 - <mc_root>/production/classifiers/ecdf_aggregated_ovr.pkl
 - <mc_root>/production/classifiers/ecdf_aggregated_ovr.meta.json
- if model-MC:
 - <mc_root>/model_mc/shared/run_XXXX/...
 - <mc_root>/model_mc/<backend>/all_metrics.csv
 - <mc_root>/model_mc/backend_ranking.csv
 - <mc_root>/production/selected_backend.json (after selection)

When model-MC reuses primary split artifacts, shared/backend run roots may link centroid/detector directories instead of recomputing them.

10.7 Success checks

- final classifier/predictor artifacts exist in production root.
- selected backend metadata exists when using selection workflow.
- metrics summaries include expected selection metric fields.
- the last biological-readiness report is `go/go_with_risks` and reviewed by humans.

10.8 Do not do this

- Do not run model before freeze.
- Do not skip biological readiness review when production artifacts changed.
- Do not claim backend superiority without model-MC summaries.

10.9 See also

- docs/theory/chapters/15-model-creation-and-validation.qmd
- packages/methvlvalidation/docs/ROLLOUT.md

11 Stage: Post-model Validation

11.1 Purpose

Measure descriptive performance distributions using frozen production artifacts without retraining.

Canonical entry: Post-model validation is a step in `study_validation_lifecycle.program.json`. Legacy `methyl-validation --post-model-validation` is transitional only. See [Orchestration](#).

11.2 Entry command (canonical)

Run as part of the full lifecycle program, or re-run the lifecycle graph after freeze/model — completed actions skip via `.action_results/` manifests.

11.3 Legacy entry command

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

Optional backend override (legacy CLI):

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
--post-model-validation --post-model-backend ecdf
```

11.4 Prerequisites

- Freeze/model artifacts already built in production root.
- Valid production/project.json.

11.5 Process model

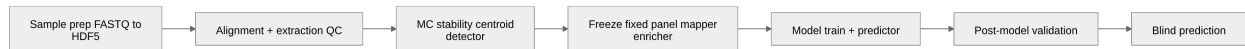


Figure 11.1: Post-model validation

11.6 Expected outputs

- `<mc_root>/post_model_validation/run_XXXX/...`
- `<mc_root>/post_model_validation/all_metrics.csv`
- `<mc_root>/post_model_validation/metrics_summary.json`
- `<mc_root>/post_model_validation/metrics_distributions_plotly.html`

11.7 Success checks

- run directories created under `post_model_validation`.
- metrics summary includes key metrics (`balanced_accuracy`, `macro_f1`, plus available calibration metrics).
- distribution plot generated.

11.8 Do not do this

- Do not treat post-model validation as model training.
- Do not compare candidate and baseline manually without fixed thresholds; use rollout comparison when promoting.

11.9 Recovery

- Resume interrupted post-model validation:

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
--post-model-validation --resume
```

11.10 See also

- `packages/methylvalidation/docs/USAGE.md`
- `packages/methylvalidation/docs/ROLLOUT.md`

12 Stage: Blind Prediction

12.1 Purpose

Run inference on newly arrived unlabeled samples using frozen model artifacts.

12.2 Important boundary

Blind-only prediction is a standalone `methyl-predictor` operation. It is intentionally outside `methyl-validation` accuracy workflows.

12.3 Entry command (standalone)

```
source .venv/bin/activate
methyl-predictor --project /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs
```

12.4 Process model

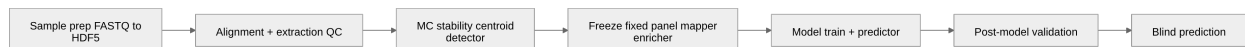


Figure 12.1: Blind prediction stage

12.5 Required config pattern

Profile `actionConfig.predictor` should reference blind/test-blind inputs and model paths resolved from frozen production project.

12.6 Expected outputs

- blind prediction report JSON
- per-sample class probabilities and predicted subgroup labels

12.7 Success checks

- predictor completes without missing model-path errors.
- expected blind report file exists.
- per-sample prediction rows match input sample count.

12.8 Do not do this

- Do not use blind-only prediction results as labeled validation accuracy claims.
- Do not run blind-only predictor settings through `methy-validation` MC paths.

12.9 Recovery

- If model path resolution fails, confirm `production/project.json` and classifier artifacts exist.
- If sample-id join fails, verify sample path naming and predictor input lists.

12.10 See also

- `packages/methylpredictor/docs/USAGE.md`
- `docs/usage/index.qmd`

Part III

Part III: Operations and Reliability

13 Artifacts and QA Checks

13.1 Purpose

Provide a single checklist for expected files at each stage and quick quality gates before proceeding.

13.2 Workflow action observability

Each workflow ACTION executed by `methy1-worker` submits a typed `output_json` to the database (validated against `schemas/tasks/*.json`). Outputs include timing (`duration_ms`), branch code (`result_code`), and paths to artifacts on shared storage.

On `/work`, look for:

- **Per-action manifests:** `{output_dir}/.action_results/{action_name}.{run_key}.json` — full typed result written by CLIs (e.g. `pipeline.dmp_select`, `pipeline.mapper`) or reconstructed by the worker collector. Schema 1.2 adds `content_key` and `hyperparam_set_id` when [CAAS](#) is enabled.
- **Sample prep timeline:** `{sampleDir}/{sampleId}.sample_prep_log.jsonl` — append-only JSONL audit for download, align, QC, extract, archive steps.
- **Workflow action timeline:** `{logRoot}/action_run_log.jsonl` — append-only log for every workflow ACTION when a log root resolves (MC runs, project output, or sample dir). Records include timing, typed `inputs/outputs`, and idempotency fields.

For `.caas/` layout, hyperparameter-set ledgers, and cross-instance reuse, see [Content-Addressed Action Store \(CAAS\)](#).

Regenerate JSON Schema artifacts after model changes:

```
source .venv/bin/activate
methy1-export-task-schemas
methy1-export-task-schemas --check # CI drift gate
```

13.3 Artifact map

Stage	Required artifacts	Gate to proceed
Sample prep	<code>{sample_id}.json</code> , <code>alignment_qc/{sample_id}.json</code> , <code>{sample_id}.extraction_qc.json</code> , <code>{chr}-CG.h5</code>	<code>guardrails.overall_pass</code> true for alignment and extraction QC
Stability	<code>stability/stable_dmps_production.csv</code> , <code>stability_summary.json</code> , <code>all_metrics.csv</code>	Stable panel exists and is non-empty
Freeze	<code>production/project.json</code> , mapper/enricher outputs	Production project exists; fixed-panel detector run completed
Model	<code>production/classifiers/*</code> , <code>production/classifiers/training_metrics.json</code> (all model backends), <code>production/predictors/*</code>	Final classifier/predictor artifacts present
Post-model validation	<code>post_model_validation/all_metrics.Dist</code> , <code>metrics_summary.json</code> , plotly report	Distribution summaries generated
Blind prediction	blind prediction report + per-sample probabilities	All input blind samples receive predictions

13.4 Handoff integrity checks



Figure 13.1: Artifact handoffs

13.5 QA checklist per stage

13.5.1 Stability

- Iteration count in metrics roughly matches configured `n_iterations`.
- Stable panel size is consistent with selected thresholds.

13.5.2 Freeze

- `production/project.json` references fixed panel detector mode.
- Mapper/enricher outputs exist for each expected comparison group.

13.5.3 Model

- Backend used is recorded and consistent with config.
- If selection used, ranking artifacts exist and chosen backend is traceable.

13.5.4 Post-model validation

- Metrics summary includes required decision metrics.
- Distribution artifacts generated and readable.

13.5.5 Blind prediction

- Prediction outputs include all requested samples.
- Operational report does not include false accuracy claims.

13.6 Escalation triggers

- Empty stable panel.
- Missing production project.
- No model artifacts after model stage.
- Repeated run failures in same step across resumed runs.

13.7 See also

- `packages/methylvalidation/docs/USAGE.md`
- `packages/methylvalidation/docs/IMPLEMENTATION.md`

14 Troubleshooting and Recovery

14.1 Purpose

Standardize response for common stage failures and reduce ad-hoc debugging.

14.2 Failure routing

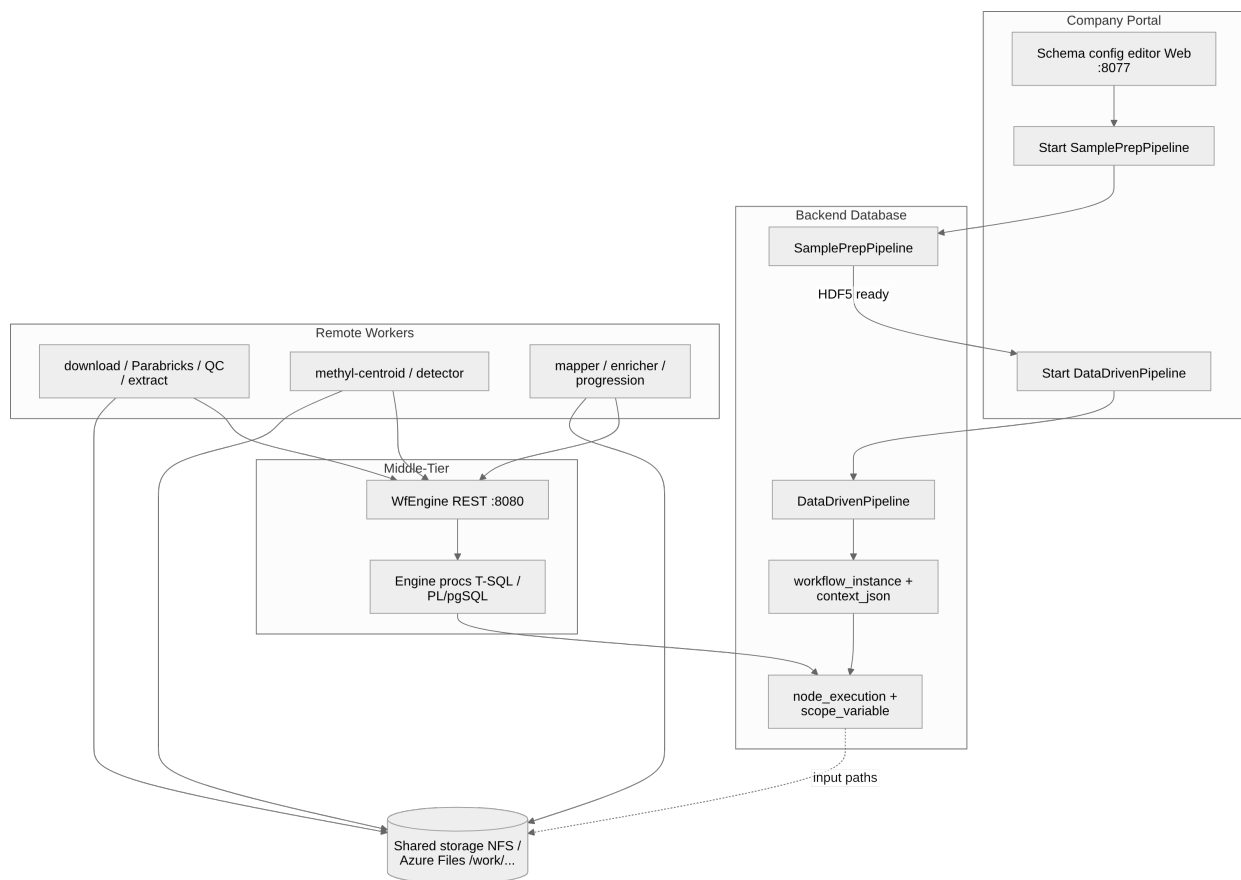


Figure 14.1: Failure routing

14.3 Common problems

Symptom	Likely cause	Recovery
<code>command not found:</code> <code>methyl-validation</code>	<code>.venv</code> inactive or install missing	Activate <code>.venv</code> , rerun <code>scripts/install_all.sh</code>
freeze fails for missing stable panel	stability not run or wrong path	Run <code>--stability</code> or set <code>freeze_stable_dmp_csv</code>
model fails due to biological review gate	<code>require_biological_review_for_model</code> set to true and not confirmed	<code>biological_review_confirmed=true</code> after review
path file-not-found across hosts	environment path differences	use <code>path_remap/--path-remap</code> <code>OLD=NEW</code>
blind predictor rejected in validation mode	blind settings passed to <code>methyl-validation</code>	run blind mode with standalone <code>methyl-predictor</code>

14.4 Safe recovery sequence

1. Validate environment and CLI availability.
2. Confirm previous-stage artifacts exist.
3. Re-run failed stage only.
4. Use resume mode where supported.
5. Re-check QA gates before advancing.

14.5 Resume examples

```
source .venv/bin/activate
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
methyl-validation --project /work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-C
```

14.6 Operational guardrails

- Keep original run directories for traceability.
- Prefer additive reruns over manual artifact edits.
- Record config overrides used in reruns.

14.7 See also

- `docs/code_first_discovery_report.md`
- `docs/reference/config-parameter-matrix.md`

15 Command Cookbook

15.1 Purpose

Provide copy-ready command sequences for the full solution lifecycle. **Prefer `methyl-workflow-run`** (see [Orchestration](#)). Legacy `methyl-validation` sequences are at the end for transitional scripts.

15.2 1) Full staged run (canonical — DomainProgram)

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/study_validation_lifecycle.program \
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-}' \
  --parallel-workers 1
```

MC stability only (freeze-readiness gate included):

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/buffy_healthy_vs_pca/configs/buffy_mc_stability.program \
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_PCa1-5-}'
```

15.3 2) Full staged run (legacy CLI — transitional only)

```
source .venv/bin/activate
PROJECT="/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"

methyl-validation --project "$PROJECT" --stability
methyl-validation --project "$PROJECT" --freeze
methyl-validation --project "$PROJECT" --model
methyl-validation --project "$PROJECT" --post-model-validation
```

15.4 3) Backend comparison and selection (legacy CLI)

```
source .venv/bin/activate
PROJECT="/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"

methyl-validation --project "$PROJECT" --model-mc --model-mc-all
methyl-validation --project "$PROJECT" --select-best-model --model-mc-all \
  --selection-metric balanced_accuracy --selection-stat median
```

15.5 4) Rollout comparison report (legacy CLI)

```
source .venv/bin/activate
PROJECT="/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"
BASE="/work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/model_mc/ecdf/metric"
CAND="/work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/model_mc/tabular_sk"

methyl-validation --project "$PROJECT" --rollout-compare \
  --baseline-summary "$BASE" --candidate-summary "$CAND"
```

15.6 5) Standalone enricher run from mapper output

```
source .venv/bin/activate
methyl-enricher \
  --input /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/mapper/all/pca_pca1/all-gene_names
  --modules \
  --library-preset cancer-extended \
  --outdir /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/enricher/all/pca_pca1
```

15.7 6) Standalone blind prediction

```
source .venv/bin/activate
methyl-predictor --project /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs
```

15.8 7) Manual rendering commands (this manual)

```
quarto render docs/usage --to html
quarto render docs/usage --to pdf
```

15.9 8) Distributed queue (legacy — one task)

Full workflow: [Distributed MethylValidation on shared storage](#) (Chapter 13). Prefer gateway workers for new deployments. One file-queue worker invocation:

```
source .venv/bin/activate
methyl-validation run-task --task /work/.../monte_carlo_runs/queue/tasks/run_0001.json
```

15.10 9) Hyperparameter search grid (example)

After stability is understood (via workflow-run or legacy CLI), use the grid driver. Details: [Optional hyperparameter search](#) (Chapter 15).

```
source .venv/bin/activate
methyl-hyperparam-search \
  --config /work/experiments/my_mc_config.json \
  --work-dir /work/experiments/hp_run1 \
  --grid '{"stability_dmp_freq": [0.6, 0.7]}' \
  -- --stability
```

15.11 Execution notes

- Always activate `.venv` before `methyl-*` commands.
- Use absolute project paths in production scripts.
- Keep command logs with timestamp and git commit hash for audit.
- New studies: start from [Orchestration \(ch.04\)](#), not legacy stage flags.

16 Distributed MethylValidation on shared storage

Status: This chapter documents the **legacy file-queue** path (`plan-runs / run-task`). New cluster deployments should use `methyl-gateway + methyl-worker` with the same DomainPrograms as local `methyl-workflow-run` — see [Deployment and distributed workflow](#) and [Orchestration](#).

16.1 Purpose

Run discovery Monte Carlo (and related queue-backed work) on **several machines** when a single host cannot finish `n_iterations` in acceptable wall time. Every worker must read and write the **same** `output_base` tree (commonly shared NFS under `/work/...`).

16.2 Prerequisites

- One **MonteCarloConfig** (dedicated `mc.json` or profile `actionConfig.validation` merged at plan time) with `output_base` on shared storage
- The same virtual environment and `methyl-*` CLIs on each worker
- R/W access to: `output_base/<project_name>/monte_carlo_runs/` and the `queue/` subtree created by planning

16.3 Layout (under `monte_carlo_runs/`)

Path	Role
<code>queue/mc_config.json</code>	Snapshotted config for workers
<code>queue/plan_runs.json</code>	Planned runs and task JSON paths
<code>queue/tasks/run_####.json</code>	One task = one <code>run-task</code> invocation
<code>queue/queue_manifest.jsonl</code> / <code>queue/commands.sh</code>	Optional export for a job scheduler
<code>run_####/queue_task_status.json</code>	Worker completion status
<code>run_####/queue_local_step_timings.json</code>	Merged by <code>aggregate-results</code> into <code>step_timings.csv</code>

Full path reference: `packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md`.

16.4 Command sequence

1) **Plan** (one process; creates run dirs and task JSON, no heavy steps):

```
source .venv/bin/activate
methyl-validation plan-runs --config /work/experiments/my_mc_config.json --overwrite
# or: methyl-validation plan-runs --project /work/projects/prostate-cancer/configs/project_Hea
```

`--overwrite` only replaces `queue/tasks/` and regenerates the plan; it does **not** remove existing `run_####` output directories. To delete all `run_####` trees before planning, add `--wipe-runs` (irreversible). Without `--overwrite`, completed runs (see `queue_task_status.json` + per-run task under `queue/tasks/`) are not written over; you can add iterations and only the new `run_####` work is planned. `run-task` skips pipeline work for runs already marked completed unless you pass `--force`.

2) **Export** (optional; for a central queue broker):

```
methyl-validation export-queue --config /work/experiments/my_mc_config.json
```

3) **Workers** (one job per task; repeat until the queue is empty):

```
methyl-validation run-task --task /work/projects/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_car
```

4) **Aggregate** (after successful tasks, or to refresh from a partial set):

```
methyl-validation aggregate-results --config /work/experiments/my_mc_config.json
# add --stability to run the same DMP/gene stability pass as monolithic methyl-validation --st
```

If adaptive stop settings are enabled for monolithic stability runs, remember queue workers do not coordinate global early termination on their own. Keep `stability_early_stop_enabled=false` for queue-first executions unless your scheduler supports coordinator-side cancellation logic.

Legacy: a single process may still run `methyl-validation --stability` without subcommands; behavior is unchanged.

16.5 Example paths (/work pattern)

With `output_base` pointing at shared storage, `plan-runs / export-queue` resolve `monte_carlo_runs` the same way as the monolithic CLI: `output_base / project_name / monte_carlo_runs /`.

16.6 Recovery

- Re-run a failed `run-task` for the same `--task` path.
- `queue_task_status.json` records failed vs completed.
- `aggregate-results` skips failed runs (when status files are present) and includes runs with computable per-run metrics.

16.7 See also

- `packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md` (canonical detail)
- `docs/theory/chapters/15-model-creation-and-validation.qmd` (section *Distributed Monte Carlo on shared storage*)
- [Optional hyperparameter search](#) (Chapter 15; search trials can use the same shared storage)

17 Deployment and Distributed Workflow

17.1 Purpose

Deploy the **control plane** (database, REST gateway) and **GPU workers** that execute SamplePrep and validation workflows. This chapter consolidates the dual-backend story (Azure SQL and PostgreSQL) and links to detailed runbooks.

Deep dive: [docs/deployment/production_runbook.md](#)

Worker protocol: [workers/WORKER_PROTOCOL.md](#)

REST contract: [contracts/openapi.yaml](#)

17.2 Architecture

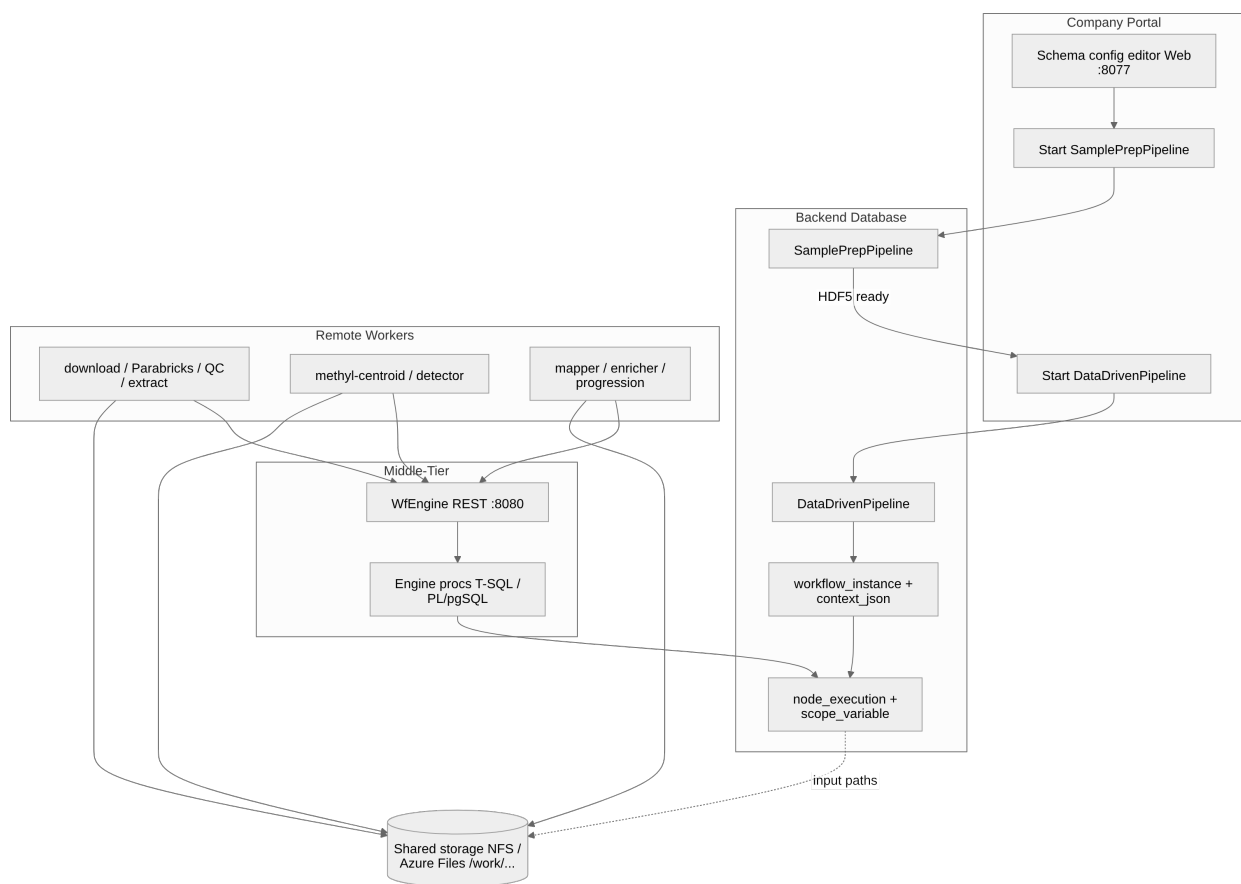


Figure 17.1: Distributed runtime

Component	Role
Database	Stores workflow definitions, instances, scope, task queue
Portal SQL (<code>portal.sp_*</code>)	Middle-tier orchestration; starts instances from study metadata
Gateway (<code>methyl-gateway</code>)	REST API for workers (poll/submit) and admin (deploy definitions)
Workers (<code>methyl-worker</code>)	Execute capabilities (Parabricks, methyl-qc, detector, ...)
/work storage	Shared FASTQs, BAMs, HDF5, MC outputs

The portal does **not** call the gateway for routine worker dispatch in production; workers poll the gateway directly. Operator/admin routes may require Entra ID JWT when `GATEWAY_REQUIRE_ENTRA=1`.

17.3 Greenfield checklist

Use this ordered checklist once per environment:

1. Mount shared storage at `/work/epimethyl` (or study-specific `/work/<disease>/`)
2. Deploy database schema (PostgreSQL or Azure SQL — see below)
3. Run portal SQL scripts (`portal_resource_profile.sql`, `portal_workflow_api.sql`, cluster security if used)
4. Seed action catalog: `methyl-export-task-schemas && methyl-export-action-catalog && python workflow_engine/sql_mssql/seed_action_catalog.py`
5. Deploy workflow definitions: `bash scripts/deploy_workflow_definitions.sh`
6. Install and configure gateway (`deploy/env/gateway.postgres.env.example` or `gateway.mssql.env.example` → `/work/epimethyl/env/gateway.env`)
7. Enable methyl-gateway systemd + nginx TLS (see [production_runbook.md](#))
8. Provision GPU workers: [worker_provision.md](#), [gpu_worker_runbook.md](#)
9. Register workers: `scripts/register_worker.sh`
10. Smoke test: `scripts/smoke_sample_prep.sh`, `scripts/smoke_study_lifecycle.sh`

17.4 Database deploy (dual-backend)

Both backends implement the same **wf** schema contract. Choose one primary backend per environment; CI and local dev often use PostgreSQL.

17.4.1 PostgreSQL (recommended for new deployments)

Step	Command / doc
Schema scripts	workflow_engine/sql_pg/README.md
Automated deploy	<code>./workflow_engine/sql_pg/deploy_azure.sh</code> (core engine + portal DDL)

Step	Command / doc
Cluster IP binding	Optional: <code>wf_cluster_security_columns.sql</code> (run manually when using IP-bound clusters)
FOREACH support	Included in <code>08_foreach_support.sql</code>

```
export PGHOST=your-server.postgres.database.azure.com
export PGDATABASE=postgres
export PGUSER=dba
export PGPASSWORD='...'
export PGSSLMODE=require
./workflow_engine/sql_pg/deploy_azure.sh

# Optional: cluster IP binding columns (when using wf.cluster security features)
psql -f workflow_engine/sql_pg/wf_cluster_security_columns.sql
```

17.4.2 Azure SQL (legacy / phase-1)

Step	Command / doc
Script order	workflow_engine/README.md
Bundled schema	<code>workflow_engine/sql_mssql/MethylPipeline.sql</code>
FOREACH	<code>wf_sql_foreach_support.sql</code> (both backends)

```
# Example: sqlcmd against Azure SQL with scripts 1-18 from workflow_engine/README.md
```

Note: `deploy_azure.sh` applies core engine scripts plus portal DDL (`portal_resource_profile.sql`, `portal_workflow_api.sql`, `wf_drop_platform_sample_storage.sql`). Run `wf_cluster_security_columns.sql` separately when enabling cluster IP binding.

17.5 Action catalog and workflow definitions

After schema deploy:

```
source .venv/bin/activate
methyl-export-task-schemas
methyl-export-action-catalog
python workflow_engine/sql_mssql/seed_action_catalog.py

# Deploy SamplePrep + StudyValidation DomainPrograms to gateway:
export GATEWAY_URL=https://your-gateway.example.com
export GATEWAY_ADMIN_BEARER_TOKEN='...' # when Entra required
bash scripts/deploy_workflow_definitions.sh
```

Output: `workflow_versions.json` with `workflow_version_id` values for API calls.

Canonical programs (repo):

- SamplePrep: [workflow_engine/domain/fixtures/sample_prep.program.json](#)
- Study validation: check-specific `*.program.json` under `workflow_engine/domain/checks/`

Authoring language: [docs/reference/domain-program-language.md](#)

17.6 Gateway configuration

Install from repo root:

```
source .venv/bin/activate
pip install -e workflow_engine/
```

Copy and edit an example env file:

Backend	Template
PostgreSQL	deploy/env/gateway.postgres.env.example
Azure SQL	deploy/env/gateway.mssql.env.example

Install systemd unit [deploy/systemd/methyl-gateway.service](#) and nginx config [deploy/nginx/methyl-gateway.conf](#).

Key variables (see `workflow_engine/rest/connection.py`):

- `BACKEND_DB` — postgres or mssql
- `POSTGRES_*` or `AZURE_SQL_*` — connection targets
- `WF_USE_MANAGED_IDENTITY` — Azure Entra token auth for DB
- `GATEWAY_REQUIRE_ENTRA` — require JWT on admin/operator routes
- `GATEWAY_ENTRA_AUDIENCE`, `AZURE_TENANT_ID` — Entra validation

17.7 Worker configuration

Each GPU worker runs one or more `methyl-worker` systemd units (see [deploy/systemd/methyl-worker@.service](#)).

Variable	Purpose
<code>WORKER_ID</code>	Unique worker name
<code>WORKER_TOKEN</code>	Auth token from registration
<code>WORKER_CAPABILITY</code>	e.g. <code>parabricks.fq2bam</code> , <code>methyl-qc</code> , <code>pipeline.detector</code>
<code>WORKER_API_BASE</code>	Gateway URL (https://gateway.example.com)

Registration and provisioning:

- [docs/deployment/worker_provision.md](#)
- [docs/deployment/worker_node.md](#)
- [scripts/register_worker.sh](#), [scripts/provision_worker_node.sh](#)

Worker poll/submit contract: [workers/WORKER_PROTOCOL.md](#)

17.8 End-to-end operator flow

17.8.1 1. SamplePrepPipeline

```
POST /v1/studies/sample-prep/start
{
  "projectPath": "/work/epimethyl/data/project_Example.json",
  "workflow_version_id": <from workflow_versions.json>,
  "fastqStorage": { "type": "s3", "bucket": "...", "region": "us-west-2" },
  "sampleCsvs": ["/work/.../samples.csv"]
}
```

See [Sample Prep and Quality Control](#) for QC gates.

17.8.2 2. Study validation / MC

After SamplePrep **COMPLETED**, start validation via portal SQL or:

```
methy1-workflow-run \
  --program workflow_engine/domain/checks/.../configs/your_mc.program.json \
  --context-file workflow_engine/domain/profiles/your.profile.json \
  --context '{"projectPath":"/work/.../configs/project.json"}'
```

Or legacy CLI recovery: [Stage: Stability](#) (transitional methy1-validation flags).

17.8.3 3. Local workflow (no gateway)

For development without database:

```
methy1-workflow-run --program path/to/program.json \
  --context-file profile.json \
  --context '{"projectPath":"/work/.../project.json"}' \
  --stub-external
```

17.9 Related documentation

- Production runbook: [docs/deployment/production_runbook.md](#)
- Release layout: [docs/deployment/production_release.md](#)
- Portal lifecycle: [workflow_engine/docs/portal_study_lifecycle.md](#)
- SamplePrep operator guide: [workflow_engine/sql_mssql/SamplePrepFlow.md](#)
- Documentation audit: [docs/DOCUMENTATION_AUDIT.md](#)

18 Optional hyperparameter search

18.1 When to use

After a baseline workflow-run (or legacy methyl-validation path) is understood, use a controlled **outer loop** to compare candidate settings—typically **Tier A** knobs in profile `actionConfig.validation` (e.g. `stability_dmp_freq`, `n_iterations` as budget, `stability_min_balanced_accuracy`). Tuning detector and model backends (Tier B/C) is a separate, often **nested** exercise; see the theory chapter on multi-stage search tradeoffs in `docs/theory/chapters/15-model-creation-and-validation.qmd` (*Optional pipeline hyperparameter search*).

18.2 What it is

- A **read-only** scalar **objective** (J) is computed from artifacts under `monte_carlo_runs/` (mainly `metrics_summary.json`, optional `stability/stability_summary.json`).
- The methyl-hyperparam-search entry point is a small **grid** driver: for each grid point it writes `work_dir/trial_NNNN/mc_config.json` with a merged `MonteCarloConfig` and a distinct `output_base`, runs methyl-validation (e.g. with `--stability`), then scores the run.
- It does **not** use `sklearn.model_selection` CV: each evaluation is a subprocess and file read.

Formula and weights: `packages/methylvalidation/docs/HYPERPARAMETER_SEARCH.md` (full (J), `ObjectiveWeights`, constraint semantics).

18.3 Command example

Use a dedicated MC config and a work directory on shared storage if many trials:

```
source .venv/bin/activate
methyl-hyperparam-search \
  --config /work/experiments/my_mc_config.json \
  --work-dir /work/experiments/hp_run1 \
  --grid '{"stability_dmp_freq": [0.6, 0.7], "stability_min_balanced_accuracy": [null, 0.5]}' \
  -- \
  --stability
```

- JSON `null` in `--grid` omits that field so the base config value is kept.
- Results: `work-dir/search_summary.json` (per-trial rows and best feasible trial, if any).

- `--weights-json` can point to a file for `ObjectiveWeights`.
- `--baseline-summary /path/to/metrics_summary.json` enforces **rollout-style** feasibility vs a baseline (same idea as `methyl-validation --rollout-compare`); see `packages/methylvalidation/docs/ROLLOUT.md` and `packages/methylvalidation/docs/HYPERPARAMETER`
- `--dry-run` only writes trial configs and paths; no pipeline subprocesses.

18.4 Pitfalls

- **Discovery vs calibration:** settings that optimize stability panel size may not optimize NLL, Brier, or ECE; prefer a **two-stage** strategy or tight constraints, not a single ad hoc weight mix.
- **n_iterations:** treat primarily as **compute budget** to stabilize medians, not a dimension to grid-search widely without cost awareness.

18.5 See also

- `packages/methylvalidation/docs/HYPERPARAMETER_SEARCH.md`
- `packages/methylvalidation/docs/ROLLOUT.md` (promotion / dual-run thinking)
- `docs/theory/chapters/15-model-creation-and-validation.qmd` (*Optional pipeline hyperparameter search*)
- [Distributed MethylValidation on shared storage](#) (Chapter 13; parallel execution of a single config's iterations)

Part IV

Part IV: Worked Tutorials

19 Tutorial: Healthy vs. Cancer (single group or increasing stages)

19.1 What you will build

A comparison study that contrasts **one healthy control group** against either:

- **one cancer group** (binary: healthy vs. cancer), or
- **several cancer groups in increasing stages** (healthy vs. stage 1 vs. stage 2 ...).

The pipeline treats both the same way: one control pool is compared against **each** disease group. Multi-stage is just “more disease groups,” so this tutorial covers both at once.

This is a short, end-to-end walkthrough. For field-by-field detail see [Project config and layout](#); for the stability stage see [Stage: Stability](#).

19.2 Mental model (four layers)

MethylPipeline is disease-agnostic. You describe **your study** (who is healthy, who is cancer, where the samples are) in a **study manifest**; the **science parameters** (iterations, thresholds, caps) come from a **pipeline profile** shipped in the repo. You never hard-code study facts in a profile, and you never put tool parameters in a manifest.

Layer	Artifact	You edit it?
Study manifest	<code>/work/projects/<disease>/configs/manifest.json</code>	Yes — this is your study
Cohort lists	<code>/work/projects/<disease>/data/*.csv</code>	Yes — one CSV per group
Pipeline profile	<code>workflow_engine/domain/profiles/*profile.json</code>	No
DomainProgram	<code>workflow_engine/domain/checks/**/program.json</code>	No

19.3 Step 1 — Lay out cohort CSVs

Put one sample list per group under `/work/projects/<disease>/data/`. Each CSV has a **sample** header and one sample folder name per row. Names resolve against `samples_base_path` (typically `/work/samples/<name>`).

`/work/projects/prostate-cancer/data/PCaH.csv` (healthy):

```
sample
HBCST-052125-87293
HBCST-061825-52197
HBCST-051425-74294
```

Create one more for the binary case (`pca_b.csv`), or one per stage for the multi-stage case (`PCa1.csv`, `PCa2.csv`, ...). A row may also be an absolute path or a direct sample directory; bare names use `samples_base_path`.

19.4 Step 2 — Write the study manifest

19.4.1 Option A — Binary (healthy vs. one cancer group)

`/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_PCa.json`:

```
{
  "project_name": "Buffy_healthy_vs_PCa",
  "output_base": "/work/projects/prostate-cancer",
  "samples_base_path": "/work/samples",
  "controls": {
    "label": "healthy",
    "groups": [
      { "label": "all", "sample_paths": ["/work/projects/prostate-cancer/data/healthy_b.csv"] }
    ]
  },
  "diseases": {
    "label": "cancer",
    "groups": [
      { "label": "PCa", "sample_paths": ["/work/projects/prostate-cancer/data/pca_b.csv"] }
    ]
  },
  "comparisons": "control_vs_each_disease",
  "chromosomes": ["1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "X", "Y"],
  "contexts": ["CG"]
}
```

19.4.2 Option B — Increasing stages (healthy vs. stage 1 ... stage N)

To add stages, nest a `stages[]` array under a single disease parent. Each stage is its own group with its own CSV. **The parent must not carry `sample_paths` when it uses `stages`.** Stages expand to resolved labels of the form `{parent}_{stage}` — here `PCa_PCa1 ... PCa_PCa5`.

```

{
  "project_name": "Healthy_vs_PCa1-5-CG",
  "output_base": "/work/projects/prostate-cancer",
  "samples_base_path": "/work/samples",
  "controls": {
    "label": "healthy",
    "groups": [
      { "label": "all", "sample_paths": ["/work/projects/prostate-cancer/data/PCaH.csv"] }
    ]
  },
  "diseases": {
    "label": "cancer",
    "groups": [
      {
        "label": "PCa",
        "stages": [
          { "label": "PCa1", "description": "Gleason Score 3+3", "sample_paths": ["/work/p",
          { "label": "PCa2", "description": "Gleason Score 3+4", "sample_paths": ["/work/p",
          { "label": "PCa3", "description": "Gleason Score 4+3", "sample_paths": ["/work/p",
          { "label": "PCa4", "description": "Gleason Score 4+4", "sample_paths": ["/work/p",
          { "label": "PCa5", "description": "Gleason Score 4+5/5+4", "sample_paths": ["/work/p",
        ]
      }
    ]
  },
  "comparisons": "control_vs_each_disease",
  "progression_order": "explicit",
  "progression_labels": ["PCa_PCa1", "PCa_PCa2", "PCa_PCa3", "PCa_PCa4", "PCa_PCa5"],
  "chromosomes": ["1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16","17",
  "contexts": ["CG"]
}

```

A complete, annotated version of this staged manifest lives at `workflow_engine/domain/checks/pca1_5_cg/config/5-CG.json`.

19.5 Step 3 — Understand what comparisons does

`comparisons` decides which group pairs get analyzed. For a healthy-vs-stages study you almost always want `"control_vs_each_disease"`.

Value	Meaning	Use for
"control_vs_each_disease"	First control group vs. each disease group/stage	Healthy vs. stage 1, stage 2, stage 3 ... (this tutorial)
"all_pairs"	Every control group × every disease group	Multiple control strata

Value	Meaning	Use for
[{ "control_group": "all", "disease_group": "PCa_PCa1" }, ...]	Hand-picked pairs (using resolved labels)	Custom subsets

Important: "control_vs_each_disease" compares **healthy against each stage** — it does **not** run stage-vs-stage contrasts. If you omit **comparisons** entirely, the pipeline defaults to **control_vs_each_disease** for a single control group. To model the ordered stage narrative (early → advanced) after per-comparison detection, set **progression_order** / **progression_labels** as in Option B.

19.6 Step 4 — Pick a profile

The profile supplies the science knobs (iterations, thresholds, feature caps). Profiles are **process-agnostic** — study facts stay in your manifest; the profile selects which statistical branch and scope flags run.

All shipped profiles live in `workflow_engine/domain/profiles/*.profile.json` (production: `/work/epimethyl/current/runtime-bundle/domain/profiles/`). Pass one with `--context-file` or set `pipelineProfile` in `--context`.

19.6.1 Quick picks (this tutorial)

Goal	pipelineProfile
Standard binary Monte-Carlo stability (DMP + gene FeatureCuts)	<code>mc_dmp_gene_fc</code>
Multi-stage, one-vs-rest Monte-Carlo stability	<code>staged_ovr_mc</code>
Multi-stage end-to-end (stability + freeze + model)	<code>staged_full_lifecycle</code>
Progression / interpretation only (no MC)	<code>staged_progression_interpretation</code>

19.6.2 Statistical modeling modes (mc_*)

Five reusable **research axes** for Monte-Carlo stability. See also [Stage: Stability](#).

Table 19.1: Statistical modeling modes

Mode	Profile	dmp_modeling_mode	gene_modeling_mode	Use when
1	<code>mc_dmp</code>	<code>raw_pool</code>	<code>none</code>	Exploratory DMP recurrence (no FeatureCuts)
2	<code>mc_dmp_fc</code>	<code>featurecuts</code>	<code>none</code>	BA-gated DMP panel stability
3	<code>mc_gene</code>	<code>raw_pool</code>	<code>none</code>	Enricher gene recurrence (PPI hubs or Enrichr)

Mode	Profile	dmp_modeling_mode	gene_modeling_mode	Use when
4	mc_gene_fc	raw_pool	featurecuts	Gene-axis BA gate on enricher/PPI genes
5	mc_dmp_gene_fc	featurecuts	featurecuts	Full DMP + gene FeatureCuts (production dual-axis)

mc_gene and mc_gene_fc default to `actionConfig.enricher.ppi_only: true` (STRING PPI hubs). Set `ppi_only: false` in a site/profile/instance override for Enrichr pathway libraries.

19.6.3 Full profile catalog

Every profile file in the repo/runtime bundle:

Table 19.2: Shipped pipeline profiles

Profile	Detector / selection	Mapper CSV	Stability / lifecycle	Typical use
mc_dmp	discovery only	dmps-**-discovery.csv	DMP recurrence (raw pool)	DMP-axis research
mc_dmp_fc	discovery + dmp_select	dmps-**-selected.csv	DMP panel MC + FeatureCuts	BA-gated DMP panel
mc_gene	discovery only	discovery	enricher gene recurrence	Gene-axis research (PPI/Enrichr)
mc_gene_fc	discovery only	discovery	gene FeatureCuts on enricher genes	Gene-axis BA gate
mc_dmp_gene_fc	discovery + dmp_select	discovery / selected	DMP + gene FeatureCuts MC	Production binary stability
phase_a_dmp_stability	discovery + dmp_select	selected	DMP panel lock (phase A)	Two-phase: lock DMP panel first
phase_b_gene_from_stable_dmps	only stable dmps	discovery	gene FC from frozen DMPs (phase B)	Phase B after stable_dmps_production.csv
staged_ovr_mc	staged OvR + FeatureCuts	discovery	MC stability + progression	Multi-stage healthy vs. stages
staged_full_lifecycle	staged OvR + FeatureCuts	discovery	MC + freeze + model	Multi-stage end-to-end
staged_progression	discovery interpretation	discovery	mapper/enricher + progression (no MC)	Interpretation / progression only
discovery_interpretation	discovery only	discovery	single-run mapper/enricher	One-shot interpretation
structural_features	discovery only	discovery + intersections	gene×region ranked catalog	Structural / region features
full_biomarker_gene_fc	discovery + selects	selected	DMP + gene FC + biomarker filter	Biomarker-filtered gene FC
discovery_gene_featurecuts	discovery + gene FC	discovery / selected	dual-axis FeatureCuts (legacy name)	Alias → mc_dmp_gene_fc
legacy_dual	legacy inline FeatureCuts	discovery	DMP classifier panels	Historical dual-axis layout
dmp_panel_stability	discovery + dmp_select	selected	DMP panel MC	Deprecated → mc_dmp_fc
gene_enricher_stability	discovery only	discovery	enricher recurrence	Deprecated → mc_dmp

Deprecated aliases (resolve to canonical `mc_*` files in `pipeline_profiles.py`): `mc_dmp_discovery` → `mc_dmp`, `mc_dmp_featurecuts` → `mc_dmp_fc`, `mc_gene_mapper` → `mc_gene`, `mc_gene_featurecuts` → `mc_gene_fc`, `discovery_gene_featurecuts` → `mc_dmp_gene_fc`, `dmp_panel_stability` → `mc_dmp_fc`, `gene_enricher_stability` → `mc_dmp`, `buffy_mc_gene_fc` → `mc_dmp_gene_fc`.

Two-phase workflow: run Phase A with `phase_a_dmp_stability`; pass `stable_dmps_production.csv` via context `stableDmpCsv` (or `freeze_stable_dmp_csv` in validation overrides) for Phase B `phase_b_gene_from_stable_dmps`.

Parameter detail: DomainProgram language — Pipeline profiles.

19.7 Step 5 — Run it

Always activate the repo virtualenv first. You pass three things: a **program** (the workflow topology), a **profile** (via `--context-file`), and a **context** JSON that points at your manifest and names the profile.

Binary (healthy vs. one cancer group):

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/buffy_healthy_vs_pca/configs/buffy_mc_stability.program \
  --context-file workflow_engine/domain/profiles/mc_dmp_gene_fc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Buffy_healthy_vs_PCa1-5-"}' \
  --parallel-workers 1
```

Multi-stage (healthy vs. PCa1...PCa5):

```
source .venv/bin/activate
methyl-workflow-run \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/pca1_5_mc_stability.program.json \
  --context-file workflow_engine/domain/profiles/staged_ovr_mc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-"}' \
  --parallel-workers 1
```

Before a full run, you can validate the manifest and compile the program with the check harness:

```
source .venv/bin/activate
python workflow_engine/domain/checks/pca1_5_cg/check_pipeline.py \
  --program workflow_engine/domain/checks/pca1_5_cg/configs/pca1_5_mc_stability_smoke.program \
  --context-file workflow_engine/domain/profiles/staged_ovr_mc.profile.json \
  --context '{"projectPath":"/work/projects/prostate-cancer/configs/project_Healthy_vs_PCa1-5-"}'
```

In production, point `--program` and `--context-file` at the promoted runtime bundle instead of the repo, e.g. `/work/epimethyl/current/runtime-bundle/domain/...` (see [Deployment and distributed workflow](#)).

19.8 Step 6 — Find the outputs

Results land under `output_base/project_name/`. Each comparison gets its own detection folder keyed by control and disease group:

```
/work/projects/prostate-cancer/Healthy_vs_PCa1-5-CG/
  detections/all/PCa_PCa1/      # healthy vs. stage 1
  detections/all/PCa_PCa2/      # healthy vs. stage 2
  ...
  monte_carlo_runs/             # MC iterations, stability summaries
```

For a binary study the single comparison appears under `detections/all/PCa/`. Monte-Carlo stability writes recurrence panels and `stability_summary.json` under `monte_carlo_runs/`; see [Artifacts and QA checks](#) for what each artifact means and the handoff checks before the freeze stage.

19.9 Recap

1. One CSV per group under `data/`.
2. One manifest: `controls + diseases` (nest `stages[]` for increasing stages), `comparisons: "control_vs_each_disease"`.
3. Pick a profile from the full catalog (`mc_dmp_gene_fc` binary dual-axis, `mc_*` research axes, `staged_*` multi-stage).
4. `methy1-workflow-run --program ... --context-file ...profile.json --context '{"projectPath": "...", "pipelineProfile": "..."}'`.
5. Read results per comparison under `<project_name>/detections/<control>/<disease>/`.

19.10 See also

- [Project config and layout](#) — every manifest field
- [Stage: Stability](#) — profiles, modeling modes, required `actionConfig` keys
- [DomainProgram language](#) — program vs. project vs. profile vs. context
- `packages/methylutils/docs/COHORT_TREE.md` — stages and comparison-shorthand semantics
- [Command cookbook](#) — copy-ready command sequences

Part V

Part V: Hyperparameter Versioning

20 Content-Addressed Action Store (CAAS)

20.1 Purpose

Version idempotent workflow action results by **hyperparameter and input signature** on shared storage. When two workflow instances reach the same action with identical cumulative inputs, they **reuse one physical copy** instead of overwriting canonical paths.

CAAS complements (does not replace) per-output-dir manifests at `{output_dir}/.action_results/`.

20.2 Default behavior

CAAS is **enabled by default** for every project and workflow run. No environment variable or task flag is required.

20.2.1 When to disable

- **Per instance / task:** `"caasEnabled": false` in instance context or task input.
- **Worker / site opt-out:** `METHYL_CAAS_ENABLED=0` (also accepts `false`, `no`, `off`).

Disable only for debugging when you need to inspect plain files without symlinks.

20.3 Content key

```
content_key = sha256(action_revision + "|" + input_signature)
```

`input_signature` is **cumulative**: path normalization resolves symlinks, so upstream CAAS outputs affect downstream keys. Any config or upstream artifact change forks a new store entry; prior results remain under their old keys.

20.4 On-disk layout

`{project_root}` = `{output_base}/{project_name}` from the study manifest.

```
{project_root}/.caas/
  {action_safe}/{content_key}/
    <product artifacts...>
    manifest.json
  instances/{hyperparamSetId}.json
```

Path	Meaning
<code>{project_root}/.caas/{action_safe}/{content_key}</code>	Manifest
<code>{project_root}/.caas/{action_safe}/{content_key}/manifest.json</code>	The product artifacts for one action execution
<code>{project_root}/.caas/{action_safe}/{content_key}/instances/{hyperparamSetId}.json</code>	ArtifactManifestRegion (schema 1.2) with <code>content_key</code> , <code>hyperparam_set_id</code>
<code>{project_root}/.caas/instances/{hyperparamSetId}.json</code>	Ledger: <code>{action}:{run_key} → content_key</code>
<code>{output_dir}/.action_results/{action}.run_key.json</code>	Canonical path/replay manifest (may reference CAAS paths)

Canonical output paths (e.g. under `run_0001/detections/...`) become **symlinks** into `.caas/` after a successful commit.

20.5 Hyperparameter set identity

`finalize_instance_context` bakes `hyperparamSetId` from all merged `resolvedConfig_*` slices (optional label via `hyperparamSetName`). Every ACTION template receives `hyperparamSetId`; workers stamp it on manifests and append ledger entries.

Use the ledger to compare which content keys each hyperparameter set resolved to (for example Balanced Accuracy per set).

20.6 Database mirror (distributed gateway)

When the gateway and workers run against PostgreSQL or Azure SQL:

- `wf.hyperparameter_set` — set identity and merged config snapshot
- `wf.workflow_instance.hyperparameter_set_id` — links instance to set
- `wf.hyperparameter_set_action_entry` — `action/run_key → content_key`

Deploy: `workflow_engine/sql_pg/wf_hyperparameter_set.sql` (parity in `sql_mssql/`). Instance creation calls `wf_apply_hyperparameter_set`; successful task submits upsert action entries when CAAS is enabled.

20.7 Enabled actions

CAAS applies to idempotent actions: all `pipeline.*` actions plus `validation.plan_iterations`, `validation.stability`, `validation.stability_freeze_readiness`, and `validation.prepare_freeze_project`.

20.8 Skip, reuse, and force re-run

1. Worker computes `content_key` from catalog revision + validated input.
2. If a valid CAAS entry exists, canonical paths are relinked and the action is skipped (`skipped: true` in `action_run_log.jsonl`).
3. On fresh execution, product artifacts are committed to `.caas/` and canonical paths are symlinked.

Force a fresh entry (ignore skip):

```
methyly-workflow-run --program path/to/program.json --context-file profile.json --force-rerun
```

Or set `"forceRerun": true` in context / per-action input, or `METHYL_FORCE_RERUN=1`.

20.9 Observability checklist

- ☐ No opt-out is set (`caasEnabled` not false; `METHYL_CAAS_ENABLED` not 0/false)
- ☐ `{project_root}/.caas/` appears after first successful idempotent action
- ☐ `.action_results/*.json` records include `content_key` and `hyperparam_set_id` (schema 1.2)
- ☐ Second instance with unchanged upstream config reuses entries (check `skipped: true` in `action_run_log.jsonl`)
- ☐ Config change produces a new `content_key` without deleting the old store entry

20.10 See also

- [Artifacts and QA checks](#) — stage artifact map and handoff gates
- [DomainProgram language — CAAS](#)
- [Hyperparameter result versioning plan](#)
- `packages/methylDomain/methyl_domain/content_store.py` — commit, relink, ledger implementation